

Unitree / Yushu Technology (A)

**Initiate at Underperform with RMB350 PO;
Scaling the body, building the brain**

Initiating Coverage: UNDERPERFORM | PO: 350.00 CNY | Price: 451.02 CNY

Initiate at Underperform: all good, except valuation

We expect global humanoid robot (HR) unit sales to reach 1.2mn in 2030, driven by: (1) rising end-demand amid population aging and labor shortages; (2) technological improvements, particularly in AI and motion control; and (3) declining BOM costs. Although the HR industry is still awaiting its “ChatGPT moment”, with limited generalization capabilities constraining near-term adoption, Unitree has already established an early track record of profitability as a leading global supplier of HR and quadruped robots. Unitree’s leadership in quadruped robots has enabled itself to develop scalable manufacturing capabilities that are transferable to HR. We estimate a strong 57% total revenue CAGR over 2025-28, supported by growth in HR and quadruped robot sales. We initiate coverage of Unitree at Underperform with a PO of RMB350 as its current valuation looks stretched to us (43x 2027E sales, vs. peers’ average 14x). Our PO is based on DCF valuation. **Risks:** slower HR development, slower n, competition, faster than expected ASP fall, geopolitical uncertainty.

Co-work with NVDA to expedite embodied AI development

Unitree is a pioneer in the n of HR in China, with products used mainly in research, education, application development, and commercial displays. The company sold more than 33k quadruped robot units during 2023-25. Its capabilities in core components design and development, motion control, supply-chain management, and scalable manufacturing have enabled its expansion into HR. However, we believe the next stage of growth depends on narrowing the gap in embodied AI, particularly in foundation models, training data, and autonomous task execution. We expect increased investment in models, data infrastructure, and simulation, together with its collaboration with NVIDIA on H2 Plus, to support more capable on-device AI models.

57% sales CAGR in 2025-28E on solid HR shipments

We estimate Unitree’s HR shipments to increase from c.5k units in 2025 to 17.0k/31.7k/52.7k units in 2026-28, while quadruped robot shipments to reach 32.8k/44.0k/51.7k units. Broader applications and improving affordability should support adoption, with humanoid/quadruped robot ASPs declining at 17%/6% CAGRs over 2025-28E, driven by scale-up and design optimization. We estimate a 57% sales CAGR over 2025-28E.

Estimates (Dec) (CNY)	2024A	2025A	2026E	2027E	2028E
Net Income (Adjusted - mn)	95	278	730	1,212	1,720
EPS	0.3	0.8	1.8	3.0	4.3
EPS Change (YoY)	NM	191.4%	136.2%	66.0%	41.9%
Free Cash Flow / Share	0.5	1.7	1.3	1.9	3.6
Valuation (Dec)					
P/E	1,720x	590x	250x	150x	106x
EV / EBITDA*	NM	487.3x	176.0x	106.0x	73.4x
Free Cash Flow Yield*	0.1%	0.3%	0.3%	0.4%	0.8%

* For full definitions of *IQmethod™* measures, see page 37.

30 September 2026

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Stock Data

Price	451.02 CNY
Price Objective	350.00 CNY
Date Established	30-Sep-2026
Investment Opinion	C-3-9
52-Week Range	450.70 CNY-1,100 CNY
Mrkt Val / Shares Out (mn)	27,176 USD / 404.5
Market Value (mn)	182,422 CNY
Average Daily Value (mn)	NA
Free Float	8.0%
BofA Ticker / Exchange	XYAJF / SHH
Bloomberg / Reuters	688836 CH / 688836.SS
ROE (2026E)	12.3%
Net Dbt to Eqty (Dec-2025A)	-53.9%

For acronyms, see page 34

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Refer to important disclosures on page 38 to 40. Analyst Certification on page 35. Price Objective Basis/Risk on page 35.

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iQprofileSM Unitree / Yushu Technology (A)

Key Income Statement Data (Dec)	2024A	2025A	2026E	2027E	2028E
(CNY Millions)					
Sales	393	1,699	2,713	4,312	6,552
Gross Profit	225	1,027	1,494	2,274	3,275
Sell General & Admin Expense	(84)	(541)	(373)	(447)	(623)
Operating Profit	69	335	818	1,268	1,837
Net Interest & Other Income	32	34	(10)	110	118
Associates	NA	NA	NA	NA	NA
Pretax Income	101	368	808	1,378	1,955
Tax (expense) / Benefit	(5)	(90)	(78)	(165)	(235)
Net Income (Adjusted)	95	278	730	1,212	1,720
Average Fully Diluted Shares Outstanding	364	364	404	404	404

Key Cash Flow Statement Data

Net Income	95	278	730	1,212	1,720
Depreciation & Amortization	9	24	174	379	542
Change in Working Capital	76	(9)	(63)	(214)	(196)
Deferred Taxation Charge	NA	NA	NA	NA	NA
Other Adjustments, Net	12	377	0	0	0
Cash Flow from Operations	192	670	841	1,378	2,067
Capital Expenditure	(10)	(50)	(300)	(600)	(600)
(Acquisition) / Disposal of Investments	(708)	(438)	0	0	0
Other Cash Inflow / (Outflow)	4	16	0	0	0
Cash Flow from Investing	(714)	(472)	(300)	(600)	(600)
Shares Issue / (Repurchase)	885	695	5,917	0	0
Cost of Dividends Paid	0	0	0	0	0
Cash Flow from Financing	860	675	5,917	0	0
Free Cash Flow	183	620	541	778	1,467
Net Debt	(561)	(1,403)	(7,862)	(8,639)	(10,106)
Change in Net Debt	(354)	(839)	(6,459)	(778)	(1,467)

Key Balance Sheet Data

Property, Plant & Equipment	11	35	162	384	443
Other Non-Current Assets	246	954	953	952	951
Trade Receivables	21	59	67	134	172
Cash & Equivalents	566	1,419	7,878	8,655	10,122
Other Current Assets	684	741	907	1,348	1,824
Total Assets	1,528	3,209	9,967	11,472	13,511
Long-Term Debt	0	0	0	0	0
Other Non-Current Liabilities	9	48	48	48	48
Short-Term Debt	5	16	16	16	16
Other Current Liabilities	233	540	651	944	1,262
Total Liabilities	247	604	715	1,008	1,326
Total Equity	1,281	2,605	9,252	10,464	12,185
Total Equity & Liabilities	1,528	3,209	9,967	11,472	13,511

iQmethodSM - Bus Performance*

Return On Capital Employed	9.6%	12.7%	12.3%	12.0%	14.9%
Return On Equity	12.1%	14.3%	12.3%	12.3%	15.2%
Operating Margin	17.6%	19.7%	30.2%	29.4%	28.0%
EBITDA Margin	19.8%	21.1%	36.6%	38.2%	36.3%

iQmethodSM - Quality of Earnings*

Cash Realization Ratio	2.0x	2.4x	1.2x	1.1x	1.2x
Asset Replacement Ratio	1.1x	2.1x	1.7x	1.6x	1.1x
Tax Rate (Reported)	5.1%	24.5%	9.6%	12.0%	12.0%
Net Debt-to-Equity Ratio	-43.8%	-53.9%	-85.0%	-82.6%	-82.9%
Interest Cover	NA	NM	17.3x	NA	NA

Key Metrics

* For full definitions of iQmethodSM measures, see page 37.

Company Sector

Industrials/Multi-Industry

Company Description

Unitree Robotics is a global leader in high-performance general-purpose robotics, offering humanoid robots, quadruped robots, embodied AI, and core robotic components for research, industrial, public service, and consumer applications.

Investment Rationale

We have an Underperform rating on Unitree as we believe a rapid sales growth of its humanoid robot products in 2026-28E is already reflected in its current valuation, and it may see GPM fall in next 2-3 years due to competition.

Stock Data

Price to Book Value

19.7x



Contents

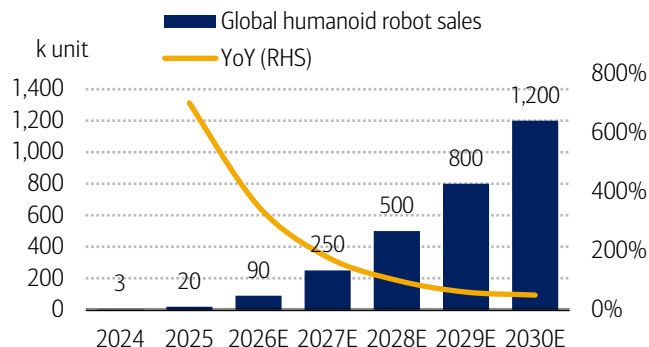
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Focus charts

Exhibit 1: BofA global humanoid robot shipments forecast

We expect annual humanoid robot shipments to reach 1.2mn by 2030

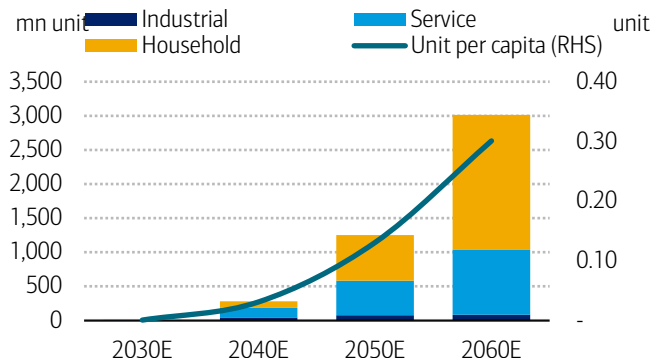


Source: Unitree, BofA Global Research estimates

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Exhibit 2: We estimate total HR UIO to reach 3.0 billion units by 2060

We expect estimate ~0.3 units per capita by 2060

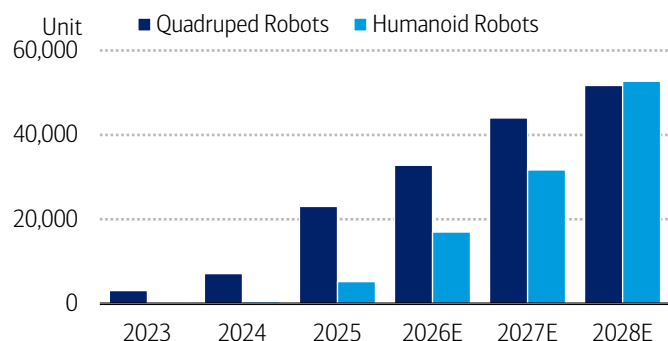


Source: BofA Global Research estimates

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Exhibit 3: Unitree's humanoid robot shipment

We expect quadruped/humanoid robot shipment CAGR of 31%/116% in 2025-28

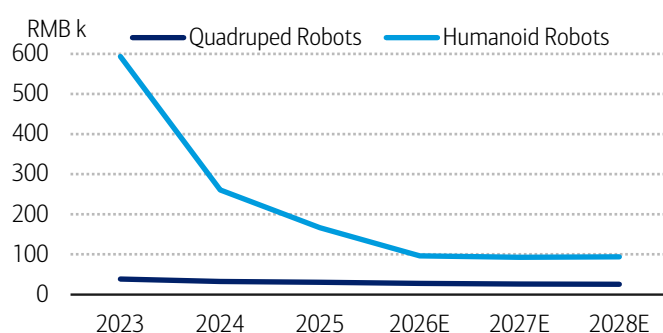


Source: Unitree, BofA Global Research

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Exhibit 4: Unitree's humanoid robot ASP

We expect the ASP of quadruped/humanoid robot to decline at 6%/17% CAGR over 2025-28

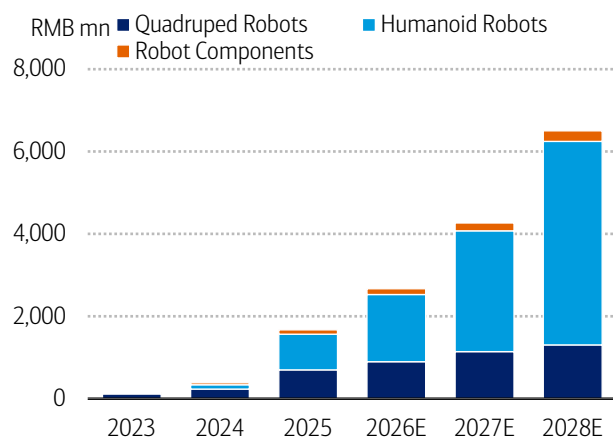


Source: Unitree, BofA Global Research

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Exhibit 5: Unitree's revenue breakdown

We expect Unitree to deliver 57% revenue CAGR in 2025-28E

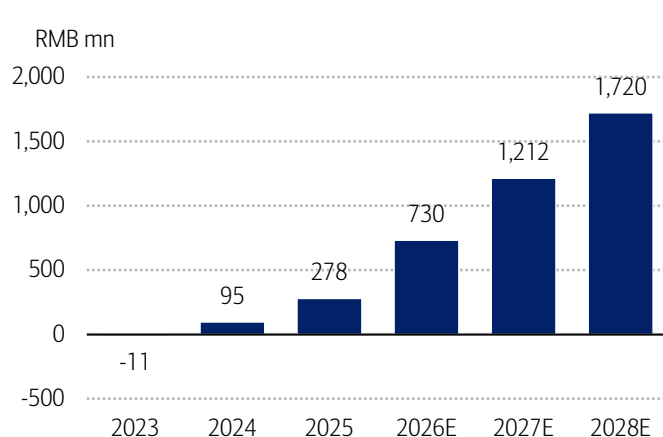


Source: Unitree, BofA Global Research estimate

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Exhibit 6: Unitree's net profit forecast

We expect Unitree to deliver 84% net profit CAGR in 2025-28E



Source: Unitree, BofA Global Research estimate

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Scaling the body, building the brain

HR: growing adoption with near-term constraints

We expect global HR shipments to grow over the next decade, driven by rising demand amid population aging and labor shortages, continued technological improvements, and declining BOM costs. We forecast global annual shipments to reach **1.2 million units in 2030 and 10.0 million units in 2035, implying an 86% CAGR over 2025-35E** (see report [Who Makes the Humanoid Robot \(in 2026\)?](#)). However, large-scale adoption remains constrained in the short term by uncertain commercial payback, insufficient robustness in contact-rich interactions, and underdeveloped embodied AI models for HR to perform multiple tasks. We therefore expect near-term adoption over the next two to three years to focus on some specific industrial applications, before gradually expanding into commercial services and, eventually, household use as performance, reliability, and affordability improve.

Early monetization and clear quadruped robot leadership

Unitree is a pioneer in quadruped robots and ranked No. 1 globally in 2024, with a 32% market share, according to IDC. Since launching Laikago in 2017, the company has developed a broad portfolio spanning consumer, research, and industrial applications, including the Go, A, and B series. Its early success in quadruped robots enabled Unitree to build capabilities in core components, motion control, supply-chain management, and scalable manufacturing. Unitree sold more than 33,000 quadruped robots during 2023-25, while the segment generated RMB698mn of revenue in 2025, accounting for 41% of total revenue. This made Unitree one of the few intelligent robotics companies to achieve **scale and profitability at an early stage**.

Narrowing embodied AI gap

Unitree has leveraged its hardware and motion-control capabilities to expand into humanoid robots through the H1, H2, G1, and R1 series. However, broader adoption will require further improvements in embodied AI models, training data, and autonomous task execution. Unitree is increasing investment in foundation models, real-world data collection, simulation, and computing infrastructure to narrow these gaps. We believe its collaboration with NVIDIA on H2 Plus could shorten the development cycle by providing access to an integrated toolchain for simulation, teleoperation data capture, and deployment. As memory capacity and bandwidth are key constraints on on-device VLA inference, we expect the compute upgrade to support the deployment of substantially larger, multi-billion-parameter models on-device.

Expect 57% sales CAGR over 2025-28

We expect Unitree to ship 17.0k/31.7k/52.7k HR units in 2026/27/28 and 32.8k/44.0k/51.7k quadruped robot units over the same period, as expanding applications and improving affordability support broader adoption. Meanwhile, we expect the ASPs of Unitree's humanoid/quadruped robots to decline by an average of 17%/6% annually over 2025-28, reflecting lower unit costs from manufacturing scale-up and design optimization. As a result, we estimate 79%/23% sales CAGRs for its humanoid/ quadruped robot businesses in 2025-28, driving a 57% CAGR in total revenue over the period. We expect HR to contribute 75% of Unitree's total sales in 2028, up from 51% in 2025.

Valuation and PO

We expect Unitree's GPM to slightly trend down in 2026-28E, reflecting ongoing price reductions to accelerate mass-market adoption and intensifying competition in China's humanoid and quadruped robot markets. We expect the company's net profit to see an 84% CAGR over 2025-28E, driven by HR shipment growth and operating leverage, partially offset by GPM normalization and higher R&D investment. Our PO of RMB350 is based on DCF valuation, reflecting Unitree's direct exposure to humanoid robot **market**, superior sales/earnings growth profile, leadership in China's HR market, and execution track record. We assume 12.0% WACC and 8% FCF perpetual growth.



Valuation

Initiate at Underperform with PO of RMB350

Our PO of RMB350 is derived using a DCF valuation methodology. We believe DCF is a suitable valuation methodology as it better captures the company's net profit, working capital management, Capex, and cash-flow generation. We believe it is reasonable given the company's early-stage n, rapid revenue growth trajectory, and ongoing R&D investment.

Exhibit 7: Valuation summary

Our PO is RMB350

Valuation methodology	Parameters	PO (RMB)	Implied 2027E P/S	Potential Downside
DCF	WACC of 12.0%	350.0	32.8x	-22.4%

Source: BofA Global Research estimates

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DCF valuation

A fair value of RMB350 is derived from our DCF calculation, with assumptions of 12.0% WACC and 8% FCF perpetual growth. Our WACC of 12.0% is based on 12.0% total cost of equity (i.e., risk-free rate of 3.5%, a risk premium of 8.5% with 1.0x equity beta) and after-tax cost of debt of 2.6%.

Within our DCF framework, we adopt a higher terminal growth rate of 8%, reflecting Unitree's direct exposure to humanoid robot n, superior sales/earnings growth profile, leadership in China's HR market, and execution track record.

Exhibit 8: Unitree's DCF forecast

We derive DCF-based valuation of RMB350

	2026F	2027F	2028F	2029F	2030F	2031F	2032F	2033F	2034F	2035F	2036F	Beyond 2036F
Sales	2,713	4,312	6,552	9,198	12,908	17,814	24,226	32,221	42,532	55,291	71,879	-
Net income	730	1,212	1,720	2,276	3,133	3,741	5,088	6,766	8,932	11,611	15,095	-
FCFF	575	643	1,174	1,731	2,438	3,782	4,585	6,060	7,937	10,292	13,155	355,260
PV of FCFF	559	558	909	1,197	1,505	2,085	2,257	2,663	3,115	3,606	4,115	111,127
Firm value (RMB mn)	133,695											
Equity value (RMB mn)	141,557											
Shares outstanding (mn shares)	404											
Value per share (RMB)	350.0											

Source: BofA Global Research estimates

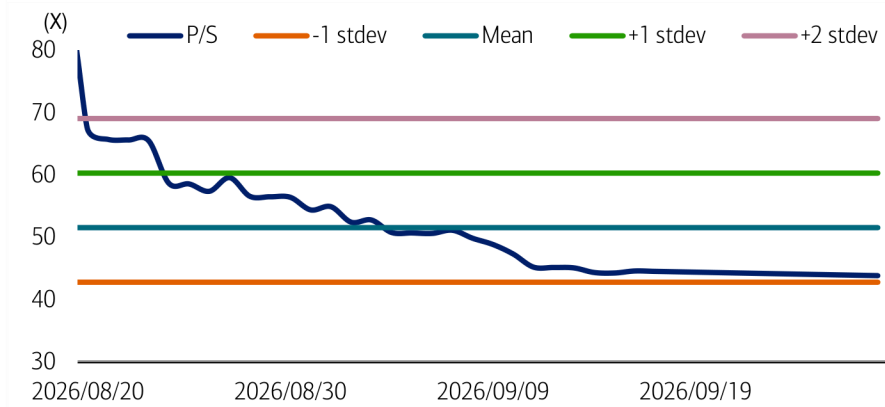
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Our PO of RMB350 implies 32.8x 2027E P/S, suggesting that the market is pricing in sustained high growth and execution visibility for Unitree to deliver in the medium term.



Exhibit 9: Unitree's 1-year forward P/S band since IPO

Historical average P/S was 52x since its listing on 19 August 2026



Source: Bloomberg

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Exhibit 10: Unitree's peers valuation comparison

Unitree's peers are trading at an average of 14x 2027E P/S

Company	Ticker	Last price (local currency)	Market Cap (\$mn)	ADTV (\$mn)	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E	25-28E	PSG 27E
Unitree*	688836 CH	451	27,208	652	68x	43x	28x	60%	59%	52%	57%	57%	57%	57%	0.8
Humanoid robot OEM															
UBTECH*	9880 HK	72.8	4,718	142	9x	6x	4x	80%	57%	40%	58%	58%	58%	58%	0.1
Humanoid robot component															
Rockchip*	603893 CH	213	13,439	394	14x	11x	9x	48%	29%	26%	34%	34%	34%	34%	0.3
Leaderdrive*	688017 CH	276	7,536	526	59x	36x	24x	53%	62%	53%	56%	56%	56%	56%	0.6
Orbbec	688322 CH	85.1	5,240	227	24x	14x	9x	53%	74%	48%	58%	58%	58%	58%	0.2
Rongtai	603119 CH	52.3	3,686	163	13x	9x	6x	35%	46%	42%	41%	41%	41%	41%	0.2
Moon's Electric	603728 CH	52.5	3,278	72	7x	6x	5x	16%	14%	15%	15%	15%	15%	15%	0.4
Zhaowei	003021 CH	69.1	2,600	68	9x	8x	7x	8%	14%	20%	14%	14%	14%	14%	0.6
Changsheng bearing	300718 CH	55.2	2,460	90	11x	9x	7x	24%	21%	19%	21%	21%	21%	21%	0.4
Keli Sensing	603662 CH	48.2	2,017	73	8x	6x	5x	15%	19%	15%	17%	17%	17%	17%	0.4
Average					22x	15x	11x	32%	35%	30%	32%	32%	32%	32%	0.4
Autonomous driving															
Horizon Robotics*	9660 HK	4.1	8,386	117	11x	7x	5x	34%	56%	50%	46%	46%	46%	46%	0.2
Momenta*	6880 HK	160	4,797	10	8x	5x	3x	65%	55%	52%	57%	57%	57%	57%	0.1
Average					10x	7x	4x	50%	56%	51%	52%	52%	52%	52%	0.1
Global Tech															
NVIDIA *	NVDA US	223	5,515,526	30,477	26x	14x	8x	65%	87%	70%	74%	74%	74%	74%	0.2
SpaceX*	SPCX US	146	1,974,438	16,464	42x	20x	11x	151%	115%	83%	115%	115%	115%	115%	0.2
Tesla*	TSLA US	358	1,411,766	19,168	13x	11x	10x	13%	15%	14%	14%	14%	14%	14%	0.8
AMD*	AMD US	608	992,333	12,783	19x	11x	8x	47%	77%	42%	55%	55%	55%	55%	0.2
Average					26x	14x	9x	69%	74%	53%	65%	65%	65%	65%	0.3
AI model															
Z.AI*	2513 HK	615	38,189	612	61x	22x	12x	488%	183%	77%	209%	209%	209%	209%	0.1
MiniMax*	0100 HK	241	10,745	315	32x	10x	4x	331%	225%	130%	218%	218%	218%	218%	0.0
Average					54x	19x	10x	409%	204%	104%	214%	214%	214%	214%	0.1
Peers' average					27x	14x	9x	88%	67%	47%	64%	64%	64%	64%	0.3

Source: BofA Global Research estimates (marked as *), Bloomberg. LC = local currency

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China HR market: towards large-scale adoption

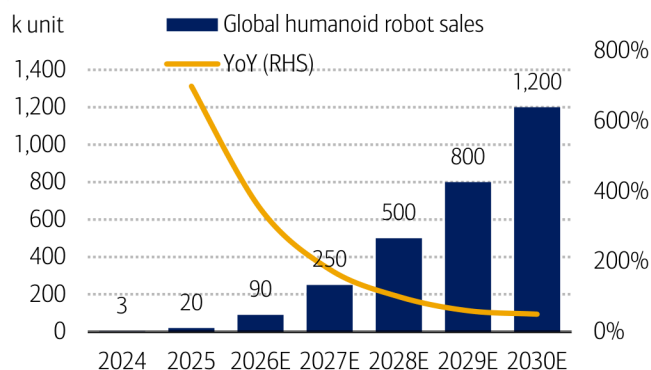
Robust shipment volume growth over coming decade

We expect HR shipments to grow strongly over the coming decade, driven by: (1) rising end-demand amid population aging and labor shortages; (2) technological improvements, particularly in AI and motion control; and (3) declining BOM costs.

We forecast global humanoid robot shipments to reach 90k units in 2026, up from 20k units in 2025. We expect annual global HR shipments to reach 1.2 million /10.0 million units in 2030/35, corresponding to an 86% CAGR over 2025-35E. We are optimistic about long-term HR demand as mass adoption begins across household and service applications.

Exhibit 11: BofA global humanoid robot shipment forecast

We expect annual humanoid robot shipments of 1.2 million by 2030

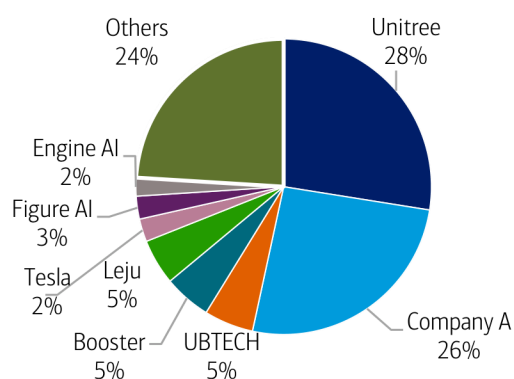


Source: BofA Global Research estimates

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Exhibit 12: 2025 humanoid robot market share (by shipment volume)

Unitree recorded 28% global market share in 2025



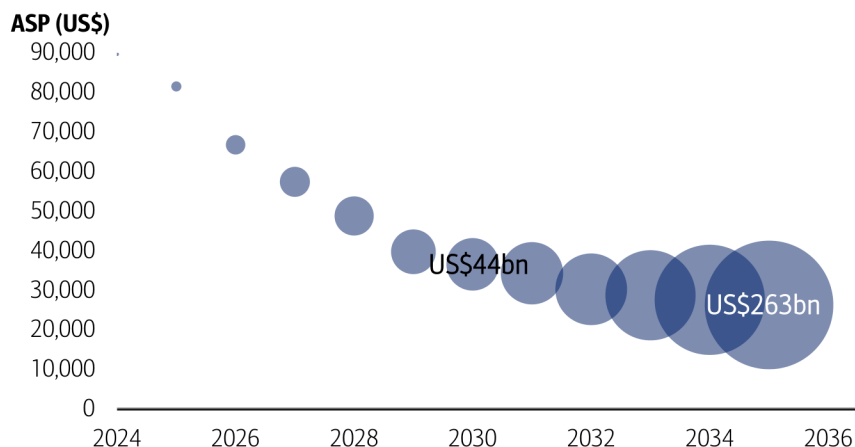
Source: BofA Global Research estimates, company reports; Company A is a private company

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We believe global HR market will reach US\$44bn/263bn by 2030/35, corresponding to a 66% CAGR over 2025-35E. This assumes ASPs of US\$37k/26k by 2030/35, vs. US\$67k in 2026, supported by continued declines in BOM and manufacturing/development costs.

Exhibit 13: Global humanoid robot market size vs. ASP

We believe the global market size of humanoid robot will reach US\$44bn/263bn by 2030/35



Source: BofA Global Research estimates

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Adoption set to accelerate in indl./commercial applications

We estimate that leading Chinese HR companies received orders worth more than RMB2bn in 2025 and RMB955mn as of mid-Sep 2026. Most orders currently come from R&D customers, including technology companies and research institutions; data collection centers supported by local governments; and manufacturers, mainly in the automotive and industrial sectors.

Manufacturing applications are usually much more difficult than R&D and data collection, as they require a higher level of reliability, stability, and consistency, which translates into high requirements for the robustness and generalization capability in manipulation.

Exhibit 14: Summary of major humanoid robot orders in China in 2025 and 2026 YTD

Major applications include R&D, data collection, and manufacturing

Date	Company	Order volume	Application	Customer
Mar-25	UBTECH	500 units	Guide service	Easyhome
Mar-25	UBTECH	20 units	Auto manufacturing	Dongfeng
Jul-25	UBTECH	RMB91mn	Auto manufacturing	Miyi (Shanghai)
Jul-25	Company A	RMB13mn	Data collection; performance	Zhuhai Embodied Intelligence Application Innovation Center
Jul-25	Company A	RMB78mn	Commercial service	China Mobile
Jul-25	Unitree	RMB46mn	Commercial service	China Mobile
Aug-25	Company A	Around 100 units	Manufacturing	Fulin Precision
Aug-25	Tiantai Robotics	10,000 units	Not specified	Shandong Future Robotics; Shandong Future Data; Gangzai Robotics
Sep-25	UBTECH	RMB30mn	Manufacturing	Miracle Automation
Sep-25	UBTECH	RMB250mn	Manufacturing	Leading domestic company
Sep-25	Astriobot	>1,000 units	Manufacturing; logistics	Shanghai Seer Intelligent
Sep-25	AI² Robotics	RMB500mn	Manufacturing; logistics	Shenzhen Huizhi IoT
Oct-25	UBTECH	RMB126mn	Data collection	Guangxi Embodied Intelligence Data Collection and Testing Center
Oct-25	UBTECH	RMB32mn	Auto manufacturing; data collection	An automotive technology company
Oct-25	Company A	Nearly 1,000 units	Manufacturing	Longcheer
Oct-25	Company A	>RMB100mn	Manufacturing	Joyson Electronics
Nov-25	UBTECH	RMB159mn	Data collection	Zigong Humanoid Robot Data Collection Center
Nov-25	UBTECH	RMB264mn	Data collection; education	Fangchenggang Humanoid Robot Data Collection and Testing Center
Nov-25	UBTECH	RMB143mn	Data collection	Jiujiang Humanoid Robot Data Collection and Training Center
Dec-25	UBTECH	>RMB50mn	Integration & secondary development	Leading Chinese LLM company
Dec-25	UBTECH	RMB60mn	Data collection	Huizhou Huiyang Humanoid Robot Data Collection Center
Dec-25	UBTECH	RMB78mn	Data collection	Hohhot Embodied Intelligence Technology Center
Dec-25	EngineAI	3,000-4,000 units	Patrol & inspection	Not specified
Dec-25	Company B	>1,000 units	Manufacturing	Zhejiang Baida Precision
Dec-25	Noetix	>1,000 units	Integration & secondary development	HCR
Jan-26	UBTECH	100 units	Manufacturing	Airbus
Feb-26	UBTECH	>RMB500mn	Auto manufacturing	BYD, Geely, Audi FAW, NIO etc.
Jun-26	UBTECH	>13,361 units	Emotional companionship	Commercial
Jun-26	Company B	RMB236mn	Logistics, inspection & services	Hechuang Energy Management
Jun-26	Company B	RMB39mn	Education	Taizhou Embodied Robotics Center (Phase II)
Jun-26	EngineAI	RMB130mn	Data collection; scenario training	Henan Qingyu Future Robotics
Sep-26	Company B	>200 units	Commercial service	Commercial
Sep-26	UBTECH	>RMB50mn	Commercial service	Overseas

Source: Company report, BofA Global Research; Company A is a private company; Company B is a private company

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Over the coming decade, we expect HR adoption to follow a three-stage trajectory: first in industrial and logistics applications, then in business services, and finally in households:

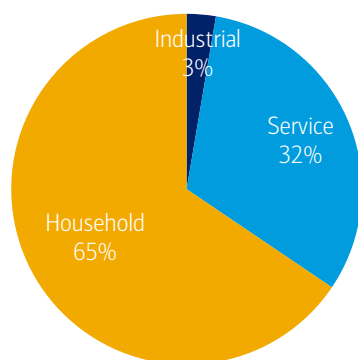
- Stage 1 (2026-29):** HRs will initially be deployed in small batches in structured or semi-structured environments, such as industrial production and logistics, mainly for material handling, assembly, sorting, and quality inspection. Small-batch testing in industrial settings has already begun (Agility Robotics, UBTECH, Leju, etc.), and we expect deployment to gradually scale over the next 3-4 years as technology matures and costs decline. This will allow HRs to accumulate real-world data for further training and model calibration.

- **Stage 2 (2029-35):** After several years of training in industrial and logistics applications, HR design and control algorithms should improve significantly. Meanwhile, increasing integration with LLMs should enable real-time human interaction. With better design and functionality, HRs will begin to see mass adoption in commercial applications, including education and business services. They will also operate in less structured environments, such as flexible manufacturing and outdoor engineering. This should support annual shipments of >1 million units.
- **Stage 3 (2036 onwards):** HRs will achieve mass adoption in general-purpose applications and highly unstructured environments, including households and elderly services. Adoption will be supported by mature functionality, smooth human interaction, and affordable production costs. At this stage, annual HR shipments could exceed 10 million units, supported by a much larger user base.

In the long run, we expect global HR units in ownership (UIO) to reach **3.0 billion**, assuming: (1) HRs replace 20% of the industrial workforce (we believe industrial and mobile robots could replace the remainder) and 50% of the service-sector workforce; (2) one HR can replace 2.5 workers in industry and 1.5 workers in services; and (3) steady-state household penetration reaches ~0.7 units per household. This equates to c.0.3 units per capita, slightly above the current level for passenger vehicles (0.2 units) but below that for smartphones (0.9 units). At steady state, we expect service/household/industrial applications to account for 65%/32%/3% of HR end-use.

Exhibit 15: Long-term application of humanoid robot breakdown

Consumer market will be the major end-market in the long run

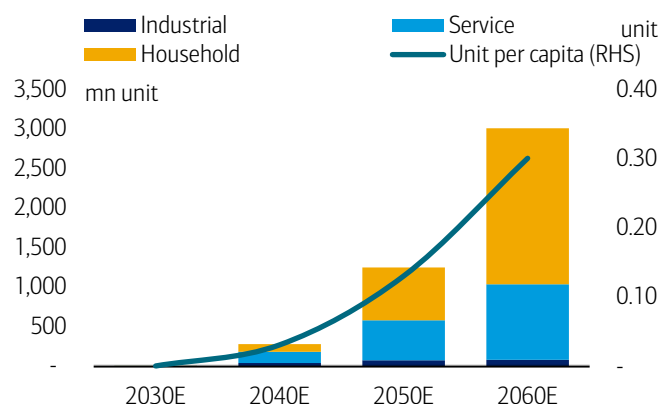


Source: BofA Global Research estimates

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Exhibit 16: We estimate total HR UIO to reach 3.0 billion units by 2060

We expect estimate ~0.3 units per capita by 2060



Source: BofA Global Research estimates

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Key challenges to large-scale monetization

We see three key challenges to the large-scale adoption of embodied AI. First, the commercial payback period must shorten, as high upfront costs and limited operating efficiency continue to constrain deployment economics. Second, robust contact interaction remains difficult, particularly in tactile sensing and dexterous manipulation. Third, the data flywheel has yet to develop: limited model capabilities constrain scaled deployment, while limited deployment restricts the accumulation of real-world data and model iteration.

Challenge #1 to monetization: cost and payback period

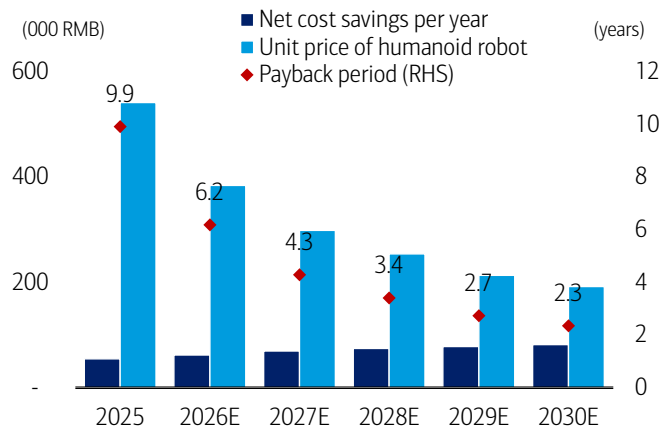
We believe the still-high cost and long payback period are among the main constraints on the large-scale use of humanoid robots in manufacturing. We estimate the payback period for a humanoid robot in China at c.6 years in 2026, reflecting the high initial purchase cost (i.e., RMB384k/unit), lower efficiency (i.e., c.45%), and lower labor costs.



Looking ahead, we estimate the payback period will shorten to 2.3 years by 2030E, mainly driven by lower ASPs and improved efficiency. In developed markets, we estimate a shorter payback period given higher labor costs, declining to **c.1 year by 2030E from 3.2 years in 2026E**.

Exhibit 17: Payback period of humanoid robot in China

We estimate payback period to shorten to 2.3 years by 2030E

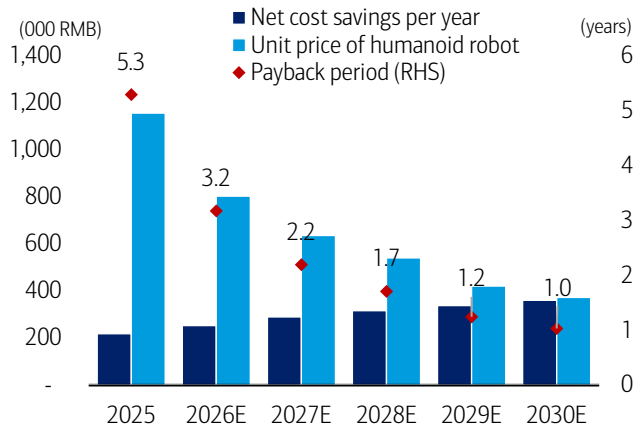


Source: BofA Global Research estimates

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Exhibit 18: Payback period of humanoid robot in developed markets

We estimate payback period to shorten to 1 year by 2030E



Source: BofA Global Research estimates

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Challenge #2 to monetization: robust contact interaction

In our view, the second bottleneck to mass adoption is the robustness of contact interaction between HRs and complex environments. HRs must sense force, friction, slippage, and deformation, and adjust their grasping strategies in real time. However, limitations in sensor accuracy, durability, and cost continue to constrain deployment in fine-manipulation use cases.

In particular, we see dexterous hands as a major technology gap, with the technology path still unsettled. The HR industry has yet to reach a consensus on degrees of freedom (DoFs), actuation, transmission, tactile sensing, and control architecture, with clear trade-offs among dexterity, payload, reliability, weight, and cost. We believe dexterous hands remain among the least mature hardware components in HRs, and technology convergence will depend on clearer use cases, a more mature supply chain, and validation through scaled deployment.

Exhibit 19: Technical route of major dexterous hand companies

Mainstream dexterous hand companies have launched products with DoF ranging from 11 to 25

Company	Robot/dexterous hand model	DoF	Transmission type
Tesla	Optimus Gen 3	22 (likely)	Tendon-driven
Unitree	Dex5-1	20 (16 active + 4 passive)	Gear/direct-drive
UBTECH	Walker S2	11	Bar linkage
Company A	OmniHand Pro	19 (12 active + 7 passive)	Bar linkage
Sharpa	SharpaWave	22	Gear/direct drive
BrainCo	Revo 2	11 (6 active + 5 passive)	Bar linkage
Leadshine	DH2015	20 (15 active + 5 passive)	Tendon-driven
PaXini	DexH13	16 (13 active + 3 passive)	Bar linkage
Inspire-Robots	RH5DG2	18 (13 active + 5 passive)	Bar linkage
Linkerbot	Linker Hand L30	25 (18 active + 7 passive)	Tendon-driven
Zhaowei	DM17	17 (all active)	Gear/direct-drive

Source: BofA Global Research, company report; Company A is a private company

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Challenge #3 to monetization: underdeveloped data flywheel

We believe the lack of a data flywheel for training embodied AI algorithms will be another key bottleneck for humanoid robots over the next 3-5 years. On the algorithm side, we note that embodied AI companies are increasingly adopting world action models (WAMs) this year, enabling greater generalization and stronger performance on long-horizon tasks. However, most embodied AI algorithms still lack autonomous generalization and adaptive response capabilities because model iteration requires large volumes of training data that are inherently scarce and costly to acquire.

Exhibit 20: Comparison of world model and VLA/VTLA model
World models generally perform well in long-process tasks

Dimension	World Model	VLA/VTLA Model
Perception Method	Physical-state modeling	Semantic understanding
Data inputs	Video sequences + physical laws	Multi-modal task data (vision-language-action)
Physical Reasoning	Strong	Relatively weak (relies on statistical patterns)
Generalization	Solves long-tail problems via synthetic simulation	Relies on task-specific data; needs retraining for new tasks
Latency	High	Low
Example	DreamerV3 (Deepmind), 1X World Model (1X Technologies), Cosmos (Nvidia)	RT-1/2 (Google), GR00T N1 (Nvidia), Helix (Figure AI), Rho-alpha (Microsoft)

Source: BofA Global Research

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Exhibit 21: Major embodied AI models for humanoid robots
Key players include robot makers, tech giants, and embodied AI start-ups

Robot company		Auto & Tech giant		Embodied AI (algorithm focus) start-up	
Company	Embodied AI model	Company	Embodied AI model	Company	Embodied AI model
Unitree	UnifoLM (UnifoLM-WMA-0, UnifoLM-VLA-0)	Tesla	Optimus AI stack (developed based on FSD & Grok)	Physical Intelligence	n0.7
Company A	GO-1	Hyundai	Gemini Robotics	Skild AI	Skild Brain
Figure AI	Helix 02	XPENG	XPENG VLA 2.0	Spirit AI	Spirit-v1.5
1X Technologies	1X World Model	Google DeepMind	RT-2, Gemini Robotics, Gemini Robotics-ER	X Square	Wall-oss-v0.5
UBTECH	Thinker	Nvidia	GR00T N1, DreamZero	GigaAI	GigaBrain-0.1
Company B	GroceryVLA, NavFoM, GraspVLA etc.	Alibaba	Qwen3.5	Dexmal	DM0
LimX	LimX VGM	ByteDance	GR-3	Generalist AI	GEN-0
Galaxea	GO Plus				

Source: BofA Global Research, company data; Company A is a private company; Company B is a private company

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The importance of high-quality real-world data has become a consensus among embodied AI developers. Egocentric data is increasingly used in pre-training because it is cheaper and available at greater scale, while high-quality real-world robot data, including teleoperation and failure-recovery trajectories, are used in post-training.

We estimate that a general-purpose embodied AI model with strong generalization capabilities would require 10-100 million hours of real-world data. Leading robotics companies now aim to collect >1 million hours of real-world data by end-2026. Third-party data service providers, such as Lightwheel, are also emerging to provide egocentric/UMI data and simulation platforms for model training.



Exhibit 22: The “data pyramid”

Real-world data come in small amount, but are of the highest quality



Source: Nvidia

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Unitree: China's humanoid/quadruped robot leader

Unitree is one of China's leading HR companies and a key global supplier of quadruped robots. It has developed its models and algorithms in house, as well as robot components including motors, reducers, dexterous robotic hands, LiDARs, and various sensors. Its product portfolio comprises: (1) **HRs** (51% of 2025 revenue), including the H1, H2, G1, R1, G1-D, and GD01 series; (2) **quadruped robots** (41% of 2025 revenue), including the AlienGo, A1, A2, As2, Go1, Go2, Go2-W, B2, and B2-W series; (3) **robot components** (6% of 2025 revenue), including the Z1 collaborative robotic arm, L1 and L2 LiDARs, Dex1-1 dexterous gripper, Dex3-1 three-finger dexterous hand, and Dex5-1 and Dex2/5 five-finger dexterous hands; and (4) **others** (2% of 2025 revenue), mainly fitness pump products.

Exhibit 23: Unitree's major business segments

Unitree provides various robotic products including humanoid robots, quadruped robots, and robot components i.e., robotic arms, dexterous hands, LiDAR, etc.

Business segment	2025 revenue contribution	2025 gross margin	Applications	Major products	Picture
Humanoid Robots	51%	63%	Assembly and delivery tasks in factory, household service, education, entertainment performances	H1, H2, G1, G1-D, R1, GD01	
Quadruped Robots	41%	57%	Industrial equipment inspection and condition monitoring, emergency search-and-rescue operations, Autonomous delivery and logistics	AlienGo, A1, A2, As2, Go1, Go2, Go2-W, B2, B2-W	
Robot Components	6%	59%	In-house supply	Collaborative robotic arm (Z1), LiDAR (L1, L2), dexterous gripper (Dex1-1), three-finger dexterous hand (Dex3-1), five-finger dexterous hands (Dex5-1, Dex2/5)	
Others	2%	71%	Fitness equipment, exhibition, etc.	Fitness pump products (PUMP, PUMP MAX)	

Source: Unitree

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HRs: front-runner in China market

Unitree ranked first globally by HR shipments in 2025, with a 28% market share, according to our previous bottom-up estimate. By application, Unitree's HRs were used in scientific research and education (74%), commercial and consumer applications (17%), and industrial applications (9%) during Jan-Sep 2025.

Exhibit 24: Comparison of major humanoid robot OEMs in China

Unitree is superior to peers on locomotion, mass production, supply chain management and cost control

Company	Unitree	Company A	Company B	UBTECH
Founded Time	2016	2023	2023	2012
Founder(s)	Wang Xingxing	Deng Taihua, Peng Zhihui	He Wang, Zhang Zhizheng	Zhou Jian
Key investors / shareholders	Meituan, Tencent, Alibaba, HongShan, Matrix Partners China	Tencent, HongShan, Hillhouse, BYD, SAIC Motor	Meituan, CATL, IDG Capital, Source Code Capital, BAIC Capital	Zhou Jian, Liuzhou Industrial Fund, Liuzhou Government Investment Fund
Key robot products / series	Humanoid: H2, H1/H1-2, G1, R1; Quadruped: Go2, A2, B2, As2	A series: A3, A2; X series: X1, X2; Genie series: G1, G2	G1, S1	Walker series: Walker S2, Walker S1
AI model	UnifoLM (UnifoLM-WMA-0, UnifoLM-VLA-0)	GO-1	GroceryVLA, NavFoM, GraspVLA etc.	Thinker
HR Shipment in 2025	5,215 units	5,168 units	1000+ units	~1,000 units
Strengths & competitiveness	Strong motion control and hardware integration; leading scale and cost competitiveness ; broad research, education and developer adoption.	Broadest product portfolio and rapid manufacturing ramp-up; balanced technology in both manipulation and locomotion .	Model-first, full-stack embodied AI approach . Its strengths include Sim2Real training, synthetic-data generation, end-to-end integration.	Industrial deployment focus and full-stack capabilities. Competitive advantages include industrial scene data, and relationships with automotive and electronics manufacturers.

Source: BofA Global Research, company report; Company A is a private company; Company B is a private company

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Unitree's humanoid product portfolio spans full-sized robots standing above 1.8m, including the H1 and H2 series, and small-to-medium-sized robots standing 1.23-1.32m, including the G1 and R1 series. The H1 and H2 series offer cutting-edge mobility and dynamic locomotion capabilities for applications such as dance performances, material handling, industrial inspection, and visitor guidance. The G1 series comprises Unitree's main commercial HRs, designed to balance affordability, scalability, and advanced motion capabilities. The lightweight R1 series is designed as an open, cost-effective platform for individual developers and educators.

Exhibit 25: Unitree's humanoid robot product portfolio

Unitree's humanoid robot products can be used in applications including performance, education, factory work, etc.

Product name	H1	G1	R1	H2
Launching time	2023	2024	2025	2025
Sales price (US\$ K)	US\$90K	US\$13.5K	US\$5.9K	US\$29.9K
Height (m)	1.80	1.32	1.23	1.82
Weight (kg)	47	35	25	70
Mobility (m/s)	3.3	N/A	N/A	N/A
Battery	Battery capacity: 0.864KWh	13 string lithium battery; battery life 2h	Lithium battery; battery life 1h	Battery capacity: 0.972KWh; battery life 3h
Degrees of freedom (DoF)	Total joint freedom: 18; each leg: 5 (Hip x3 + Knee x1 + Ankle x1), each arm: 4 (expandable)	Total: 23 (each leg: 6; waist: 1; each arm: 5)	Total: 26 (each leg: 6; each arm: 5; waist: 2; head: 2)	Total: 31 (each leg: 6; each arm: 7; waist: 3; head: 2)
Dexterous hand	Optional	/	/	Optional
Sensor Configuration	3D LIDAR + Depth Camera	3D LIDAR + Depth Camera	/	Humanoid Binocular Camera with Wide Field of View
Maximum joint torque (N.m)	360	90		360
Computing chips	Intel Core i5, i7, Nvidia Orin NX	8-Core High-Performance Processor	8-Core High-Performance Processor	Intel Core i5/i7

Source: Unitree, BofA Global Research

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H1: Unitree entered the HR market in August 2023 with the launch of the full-sized **H1**, which stands 1.8m tall, weighs 47kg, and has 18 DoF. H1 delivers strong dynamic performance and can walk and run autonomously across complex terrain, with a potential mobility of over 5 m/s.

H2: In October 2025, Unitree launched its second full-sized, general-purpose HR, the **H2**, which stands 1.82m tall, weighs 70kg, and has 31 DoF. As Unitree's latest full-sized humanoid robot, H2 can accurately reproduce highly dynamic and complex movements, making it better suited to deploying a wide range of embodied AI models across diverse work scenarios.

Exhibit 26: Unitree H1: 1.80m, 47kg, launched in 2023

Unitree's H1 humanoid robot



Source: Unitree

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G1: In May 2024, Unitree launched its first mid-sized humanoid robot, the **G1**, which stands 1.32m tall, weighs 35kg, and has 23 DoF. G1 features a more human-like structural design and a higher-DoF configuration, enabling a broader range of motion and greater dexterity. It can execute long-sequence dance routines, boxing and martial arts movements, and coordinated multi-robot operations with greater fluidity, stability, and robustness. Supported by these attributes and an attractive selling price of US\$13.5k, G1 was the world's best-selling HR in 2024 and 2025.

R1: In July 2025, Unitree launched the **R1**, a small-to-medium-sized humanoid robot that stands 1.23m tall, weighs 25kg, and has 26 DoF. Positioned as a lightweight robot, R1 can perform complex movements including running, boxing, handstands, and rolls, while also offering multimodal interaction through integrated voice and visual recognition. R1 provides individual consumers, AI developers, educational institutions, and technology enthusiasts with a high-performance, cost-effective, and highly open development platform, further accelerating technological innovation in the HR industry.

Exhibit 27: Unitree H2: 1.82m, 70kg, launched in 2025

Unitree's H2 humanoid robot



Source: Unitree

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Exhibit 28: Unitree G1: 1.32m, 35kg, launched in 2024
Unitree’s G1 humanoid robot



Source: Unitree

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Exhibit 29: Unitree R1: 1.23m, 25kg, launched in 2025
Unitree’s R1 humanoid robot



Source: Unitree

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Quadruped robots: global market leader

Unitree held a leading position in the quadruped robot market in 2024, with a 32% global market share according to IDC and a 50% share of the China market according to the Qianzhan Industry Research Institute. By application, Unitree’s quadruped robots were used in scientific research and education (32%), commercial and consumer applications (42%), and industrial applications (26%) during Jan-Sep 2025.

Unitree has built one of the industry’s most comprehensive quadruped product portfolios, comprising: (1) the A1, As2, Go1, and Go2 series for **consumer and education** applications, targeting individual consumers and developers, educational institutions, and research organizations; and (2) the A2, B1, B2, and AlienGo series for **industrial** applications, including logistics delivery, emergency response and fire rescue, factory inspection, power and energy infrastructure inspection, and security patrol.

Exhibit 30: Unitree’s quadruped robot product portfolio								
Unitree’s quadruped robot products can be used in applications including autonomous delivery, emergency search-and-rescue operations, and industrial inspection								
Application	Consumer/Education				Industrial			
Product name	A1	Go1	Go2	As2	AlienGo	B1	B2	A2
Launching time	2020	2021	2023	2026	2019	2022	2023	2025
Sales price	/	US\$2.7-3.5K	US\$1.6-4.5K	/	US\$50K	US\$100K	US\$100K	US\$100K
Stand size (cm)	50 x 30 x 40	64.5×28×40	70×31×40	72×37.8×45.7	65×31×60	112.6×46.7×63.6	109.8×45×64.5	82×44×57
Weight (kg)	12	12	15	18	21.5	50	60	42
Max mobility (m/s)	3.3	4.7	3.7	3.7-5	>1.5	1.2	> 6	5
Endurance (h)	2.5	/	2-4	2-4	2.5-4.6	2-3	4-6	5
Maximum joint torque (N.m)	33.5	35	45	95	/	/	360	180
Maximum payload (kg)	5	10	12	15	13	20	> 40	25 (ideally 35)
Degrees of freedom (DoF)	12	/	/	12	12	/	/	12

Source: Unitree, BofA Global Research

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Unitree entered the legged robotics industry in 2017 with the launch of **Laikago**, its first electrically actuated quadruped robot. In 2019, the company launched **AlienGo**, its first quadruped robot for **industrial** use. Priced at US\$50k, AlienGo features an upgraded power system and integrated body architecture. It can reach speeds of up to 1.5 m/s, traverse slopes of up to 25°, and operate for up to 4.6 hours under unloaded conditions. Its highly robust structural design can withstand substantial loads from running, jumping, collisions, and falls, enhancing operational reliability in complex environments. Unitree subsequently launched its next-generation industrial-grade quadruped robots, **B1** in 2022 and **B2** in 2023. Priced at US\$100k, these models can operate stably across challenging, rugged terrain while delivering superior mobility and long endurance under heavy payloads. In 2025, Unitree launched **A2**, its latest-generation industrial quadruped robot, for logistics support, industrial inspection, emergency response, and disaster relief. A2 offers exceptional terrain adaptability under extreme operating conditions and up to 5 hours of runtime. With a 25 kg payload, it provides a range of 12.5 km and can overcome obstacles up to 1m high and slopes of up to 45°, making it well suited to demanding industrial and field operations.

Exhibit 31: AlienGo sales at US\$50k

Unitree AlienGo quadruped robot



Source: Unitree

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Exhibit 32: B2 sales at US\$100k

Unitree B2 quadruped robot



Source: Unitree

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Exhibit 33: The latest-gen industrial robot

Unitree A2 quadruped robot



Source: Unitree

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Unitree launched **A1**, its first quadruped robot for **consumer and educational** use, in 2020. Designed for broader adoption, the cost-effective A1 supports fundamental locomotion and dynamic movements, including walking, running, and dancing. It measures 50 × 30 × 40 cm when standing and weighs 12 kg with the battery installed. A1 can reach a maximum speed of 3.3 m/s, deliver maximum joint torque of 33.5 N·m, and operate for up to 2.5 hours under unloaded conditions. The A1 series combines agility, affordability, and an open architecture, making it a popular platform for research, education, and autonomous robotics development. Unitree launched its consumer-grade quadruped robots **Go1** in 2021 and **Go2** in 2023. These models can perform dance routines and support obstacle detection and avoidance, voice-based interaction, and intelligent follow-me functionality. In 2026, Unitree launched **As2**, its latest-generation quadruped robot for **consumer and educational** use. As2 demonstrates strong environmental adaptability and mobility across challenging terrain, including ditches, snowfields, forests, rocky roads, and other harsh operating environments. Designed for inspection, research, and educational applications, As2 supports an unloaded runtime of 2-4 hours and a range of 13 km with a 15 kg payload. It is well suited to search and inspection tasks in demanding field environments, as well as educational applications.

Exhibit 34: The first consumer/education quadruped robot

Unitree A1 quadruped robot (launched in 2020)



Source: Unitree

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Exhibit 35: Go2: consumer-grade robot

Unitree Go2 quadruped robot (launched in 2023)



Source: Unitree

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Exhibit 36: The latest-gen consumer/education quadruped robot

Unitree As2 (launched in 2026)



Source: Unitree

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Full-stack provider with in-house hardware and software

Unitree is one of the few robotics companies that supplies key robot hardware components in-house – including dexterous hands, servo actuators, and LiDAR – and develops its own software systems, comprising the robot’s “brain” (embodied AI algorithms) and “cerebellum” (motion-control software). We believe this vertical integration across hardware and software strengthens supply-chain stability, accelerates technology iteration, and meaningfully lowers manufacturing costs, giving Unitree an advantage over peers in both product performance and cost competitiveness.

Key hardware technologies**Dexterous hand**

Since May 2024, Unitree has developed a series of dexterous hands for humanoid robots, including Dex1-1, Dex3-1, Dex5-1, and Dex2/5. The flagship Dex5-1 features 20 DoF, enabling lateral finger movement and significantly improving grasp stability and reliability. It also supports further development and customization of tactile-sensing algorithms. Key specifications include a maximum grasping load of 4.5 kg, four-finger lateral swing of $\pm 22^\circ$, and fingertip force of 10 N.

Exhibit 37: Dex5-1: 20 DoF and 4.5kg grasping load

Unitree Dex5-1 dexterous hand



Source: Unitree

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Exhibit 38: Dex2/5: 10 DoF and 10k fatigue cycles

Unitree Dex2/5 dexterous hand (compatible with G1/R1 models)



Source: Unitree

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Servo actuator

Unitree has developed high-power-density servo actuators in-house for its robot joint motors, enabling precise power-output control and highly efficient energy conversion. The actuators combine high load-bearing capacity with rapid transient response, supporting millisecond-level torque adjustment and over-temperature protection. These capabilities help ensure stable, reliable operation during dynamic and demanding motion tasks.

LiDAR

Unitree has developed LiDAR technology in-house, enabling it to manufacture compact, low-cost, and highly reliable products for robotics applications. The company's L1/L2 series provides omnidirectional, ultra-wide-field-of-view scanning with minimal blind spots, supporting high-precision environmental perception, robust localization, and autonomous robot navigation. The products also resist interference from indoor and outdoor lighting, helping to ensure stable ranging and mapping performance.

Exhibit 39: L1: compact size with ultra-wide angle

Unitree bionic 4D Lidar L1



Source: Unitree

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Key software technologies

Unitree has developed its robotics software systems in-house, comprising: 1) the **cerebellum**, which controls robot motion; and 2) the **brain**, which enables robots to perceive external environments and perform human-like decision-making, cognitive reasoning, and task planning. Continuous over-the-air (OTA) updates allow Unitree to enhance the adaptability of these systems over time.

Cerebellum (motion-control algorithms) comprises technologies for robot motion planning and control. Unitree's cerebellum uses deep reinforcement learning (DRL) and large-scale parallel simulation to improve locomotion in unfamiliar environments. It tightly integrates multimodal sensory inputs from LiDAR, vision systems, inertial measurement units (IMUs), and force sensors, creating a closed-loop framework that combines multimodal perception with real-time motion control.

Brain (perception and reasoning algorithms) serves as the robot's core intelligence system, enabling it to perceive and understand its surroundings, conduct cognitive reasoning, make autonomous decisions, and plan tasks. Unitree trains its robotic brain on extensive real-world data collected from physical robots and combines this data with its proprietary embodied AI models. We believe this approach enables robots to better understand natural-language instructions and environmental context, allowing them to generate appropriate actions and behavioral strategies.

Key embodied AI models

In September 2025, Unitree launched its open-source **World Model Action (WMA) model**, UnifoLM-WMA-0. The model incorporates an explicit world model that captures the physical dynamics governing interactions between robots and their environments, providing a unified cognitive and predictive foundation for robotic decision-making. It supports 10-20 rounds of multi-step interactive reasoning and simulation, enabling continuous task execution across diverse scenarios within the learned world model.

In January 2026, Unitree launched its open-source, general-purpose **Vision-Language-Action (VLA) model**, UnifoLM-VLA-0. Through continuous training on large-scale, robot-generated data, the model evolves from simply understanding images and text into an embodied AI brain with real-world reasoning and manipulation capabilities. This improves the robot's ability to perform and adapt to a range of tasks.

We believe Unitree has strong growth potential in embodied AI model capabilities, supported by its planned RMB2.0bn investment in embodied AI model R&D, representing nearly half of the proceeds from its proposed A-share offering. Although Unitree entered the embodied AI model market later than some peers, we do not view first-mover advantage as decisive at this stage, as the industry's technology stack and development path remain far from settled.

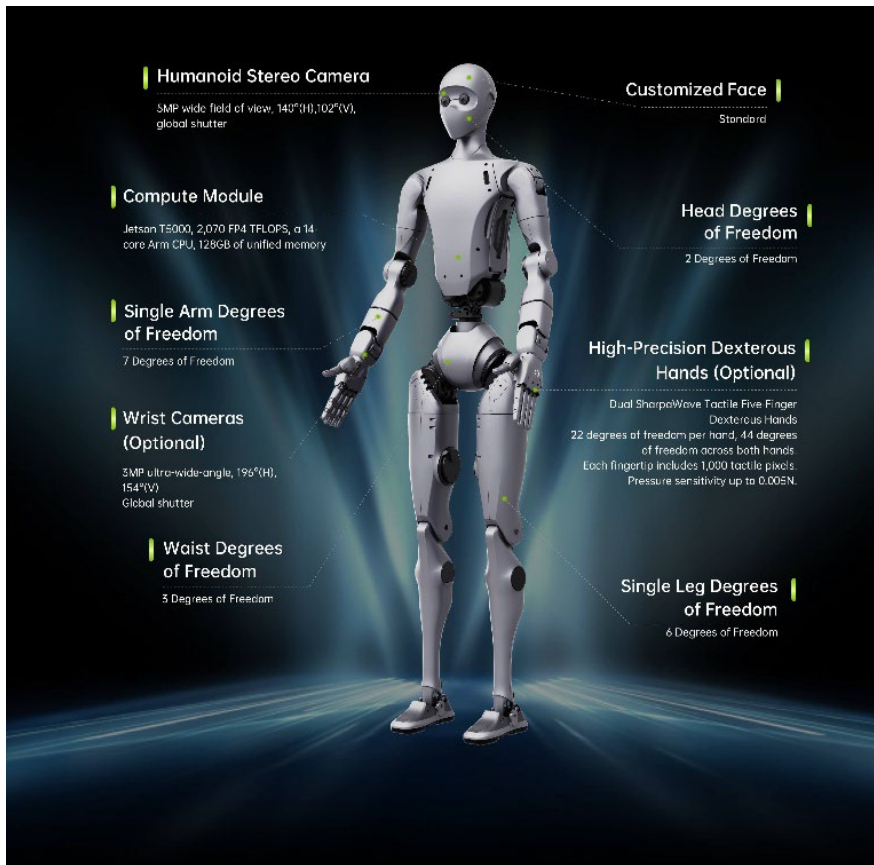
Collaboration with NVIDIA

On 1 June 2026, NVIDIA launched an open humanoid reference design under Isaac GR00T at GTC Taipei. The platform combines Unitree's H2 Plus body (c.1.8 m, c.70 kg, 31 body DoF, 120 N·m arm torque, 360 N·m leg torque, and a 7 kg rated/15 kg peak payload), NVIDIA's Jetson AGX Thor T5000 compute module, and the Isaac software stack. Singapore-based Sharpa also supplies 22-DoF, five-finger hands with more than 1,000 tactile pixels per fingertip. Launch adopters include Ai2, ETH Zürich, the Stanford Robotics Center, and UCSD, while NVIDIA Research is also deploying the platform internally. Supply through Unitree is scheduled to begin at end-2026; pricing has not been disclosed.



Exhibit 40: Unitree H2 Plus built on Isaac GR00T

Ultra-Powerful Onboard Compute for real-time sensor processing and robot inference



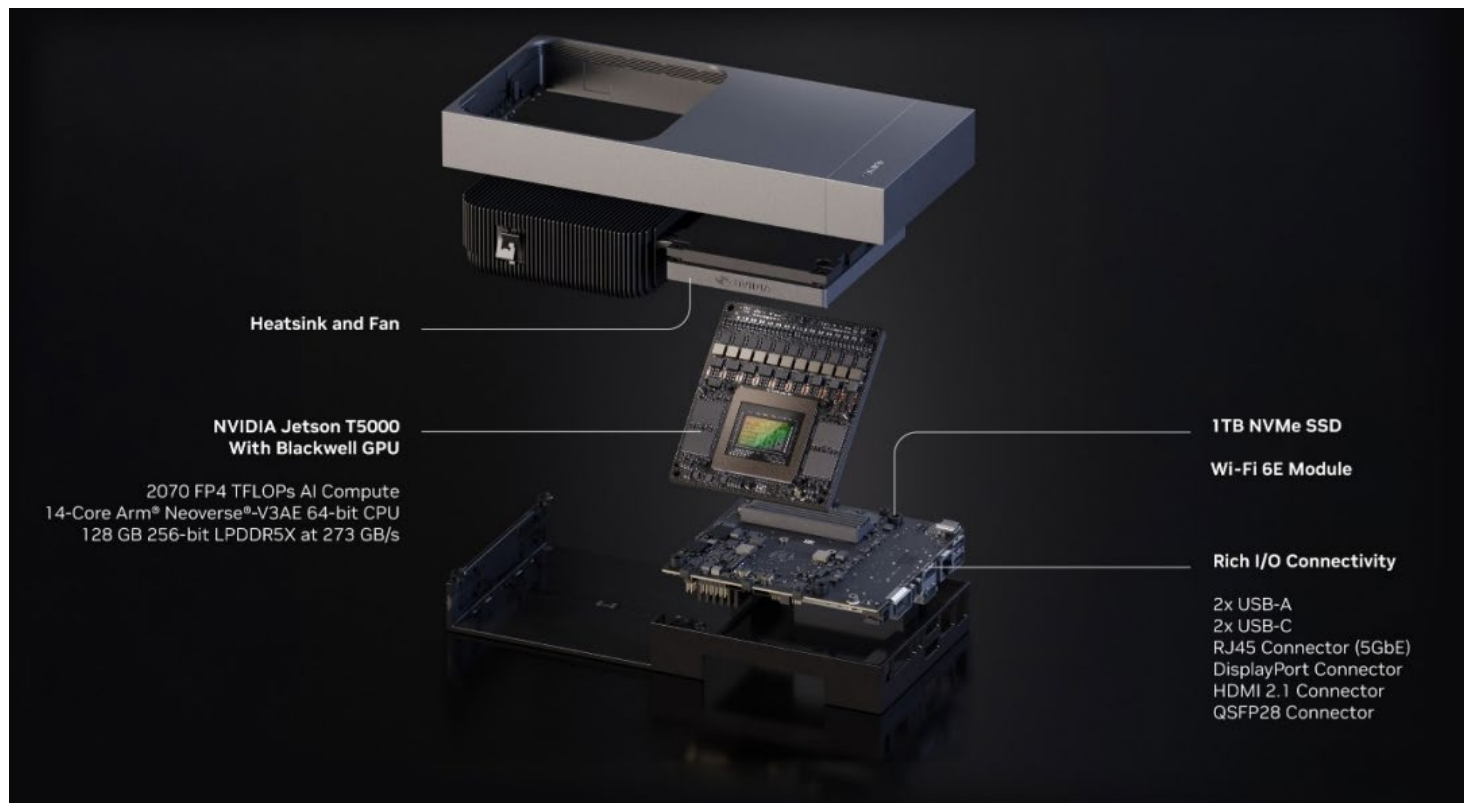
Source: Unitree

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By mid-2026, **NVIDIA** maintains a dominant position in high-end compute for HRs through its Jetson Orin/Thor series, which is widely adopted by leading companies including Boston Dynamics, Unitree, and UBTECH. Beyond NVIDIA, **Intel** and **Qualcomm** have also launched processors/systems-on-chip (SoCs) for HRs, while Tesla uses its internally developed HW4.0/AI5 chips for its Optimus HRs.

Exhibit 41: Nvidia's Jetson T5000 (flagship model of Jetson Thor)

Jetson T5000 has 2,070 FP4 TOPS (or c.1,000 INT8 TOPS) computing power



Source: Nvidia

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For Unitree, we expect the primary benefit to be that Isaac significantly reduces the need to build simulation, teleoperation data-capture, and deployment middleware from scratch. This should be particularly valuable as the company seeks to catch up with peers on the “brain” front. The other clear benefit is greater edge-compute headroom: compared with the previous Jetson Orin NX, Jetson T5000 increases memory capacity from 16.0 GB to 128.0 GB and memory bandwidth from 102.4 GB/s to 273.0 GB/s. Separately, NVIDIA claims that Jetson Thor delivers up to 7.5x the AI compute and 3.5x the energy efficiency of Jetson AGX Orin. As memory capacity and bandwidth are key constraints on on-device VLA inference, we expect the upgrade to support substantially larger, multi-billion-parameter models on-device. However, actual deployment will depend on model architecture, quantization, inference latency, and real-time control requirements.

Expect 57% sales CAGR in 2025-28

Unitree's sales rose 333% YoY to RMB1.7bn in 2025, primarily driven by strong shipment growth in quadruped robots (up 223% YoY to 23,037 units) and humanoid robots (up 1,166% YoY to 5,215 units). In 1H26, revenue growth moderated to 49% YoY, with revenue reaching RMB1.2bn, reflecting a higher base, easing industry enthusiasm after the rapid expansion phase.

Looking ahead, we expect humanoid and quadruped robots to drive Unitree's sales growth, supporting a 57% sales CAGR over 2025-28E:

- **HRs:** We expect Unitree to ship 17.0k/31.7k/52.7k HRs in 2026/27/28, driven by: (1) expanding applications across industrial, commercial, and consumer use cases; (2) customer-base growth as humanoid robots progress from pilot projects to broader deployment; and (3) improving affordability, as lower ASPs and manufacturing costs support adoption. We expect Unitree's humanoid robot ASP to decline by an average of 17% annually over 2025-28, supported by manufacturing scale-up, optimization, and product redesign. We believe these cost reductions will accelerate customer adoption and support large-scale n of humanoid robots.

Accordingly, we forecast a 79% sales CAGR for Unitree's HR business over 2025-28. We expect humanoid robots to contribute 60%/68%/75% of total sales in 2026/27/28, up from 51% in 2025.

- **Quadruped robots:** We expect Unitree to ship 32.8k/44.0k/51.7k quadruped robots in 2026/27/28, driven by broader adoption across education and research, industrial inspection, and consumer applications. We expect its established product portfolio, expanding customer reach, and improving price-performance ratio to support steady shipment growth. We expect Unitree's quadruped robot ASP to decline by an average of 6% annually over 2025-28, reflecting lower unit costs from economies of scale and greater use of shared components across models.

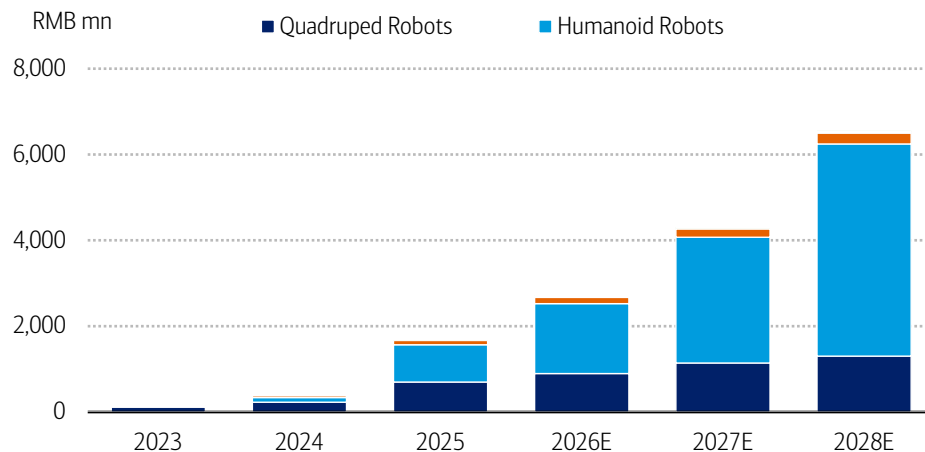
Accordingly, we forecast a 23% sales CAGR for Unitree's quadruped robot business over 2025-28. We expect quadruped robots to contribute 33%/26%/20% of total sales in 2026/27/28, down from 41% in 2025, as growth trails the hyper-growth humanoid robot business.

- **Robot components:** We expect a 35% sales CAGR for Unitree's robot components over 2025-28, driven by broader adoption among external customers of dexterous hands, collaborative robotic arms, LiDAR sensors, and joint motors. Accordingly, we expect robot components to contribute 5%/4%/4% of total sales in 2026/27/28, broadly stable vs. 6% in 2025.



Exhibit 42: Unitree's revenue breakdown

We expect Unitree to deliver 57% of sales CAGR in 2025-28



Source: Unitree, BofA Global Research estimate

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Expect 84% net profit CAGR over 2025-28

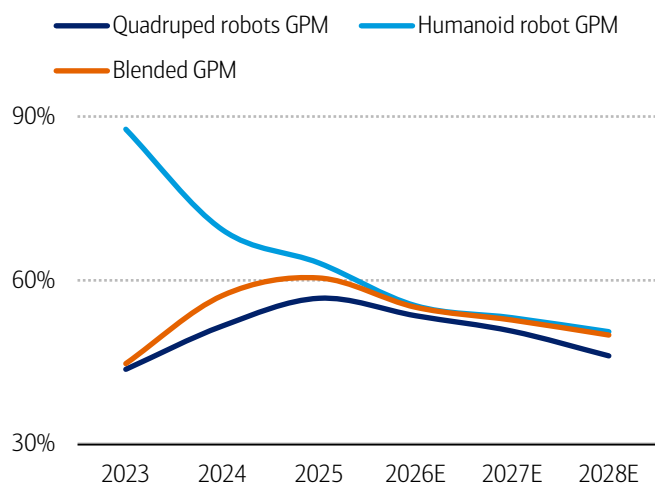
By segment, Unitree's humanoid robot GPM was 63.2% in 2025, above the 56.7% GPM for quadruped robots, mainly reflecting higher ASPs, a more premium product mix, and greater technological complexity. Humanoid robot GPM declined from 87.7% in 2023 to 69.3% in 2024 and 63.2% in 2025, owing to a sales-mix shift toward the lower margin G1 model and strategic price cuts enabled by continued cost reductions. Quadruped robot GPM rose from 43.7% in 2023 to 51.7% in 2024 and 56.7% in 2025, driven by lower unit costs from production and procurement efficiencies and a favorable product-mix shift toward higher-margin industrial models such as the B2 and B2-W.

Unitree's blended GPM increased from 44.8% in 2023 to 60.4% in 2025, driven by a rising revenue contribution from the higher margin HR business. In 1Q26, blended GPM fell 2.3ppt YoY and 3.4ppt QoQ to 58.4%, before declining a further 5.3ppt YoY and 3.8ppt QoQ to 54.6% in 2Q26. In our view, this potentially reflected price reductions and a shift toward more affordable HR models, such as the G1.

Looking ahead, we expect humanoid robot GPM to moderate to 55.4%/53.2%/50.6% in 2026/27/28, extending the downward trend seen in 1H26 as ongoing price reductions support mass-market adoption amid fiercer competition in China's HR market. We also expect quadruped robot GPM to decline to 53.5%/50.7%/46.2% in 2026/27/28, reflecting continued pricing adjustments in response to industry competition. Accordingly, we expect Unitree's blended GPM to gradually moderate from 60.4% in 2025 to 55.1%/52.7%/50.0% in 2026/27/28.

Exhibit 43: Unitree's GPM trend

We expect its blended GPM to moderate to 50.0% in 2028

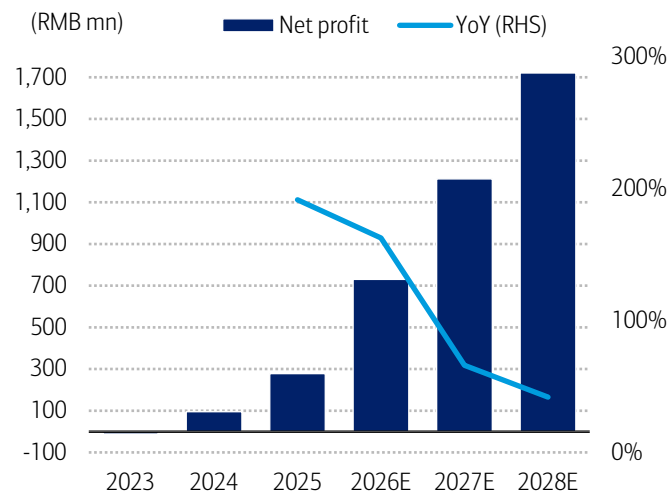


Source: Unitree, BofA Global Research estimate

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Exhibit 44: Unitree's net profit forecast

We expect Unitree's net profit CAGR at 84% over 2025-28



Source: Unitree, BofA Global Research estimate

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Exhibit 45: Unitree's revenue breakdown by segment

We expect humanoid robots to contribute 75% of Unitree's total sales by 2028

Sales (RMB mn) classification	2023	2024	2025	2026E	2027E	2028E
Quadruped Robots	119	231	698	895	1,141	1,304
Humanoid Robots	3	107	868	1,632	2,933	4,941
Robot Components	27	45	104	142	191	256
Other main business	8	5	7	20	22	24
Others	2	5	23	24	26	27
Total Sales	159	393	1,699	2,713	4,312	6,552
Sales YoY	2023	2024	2025	2026E	2027E	2028E
Quadruped Robots	-	93%	203%	28%	27%	14%
Humanoid Robots	-	3516%	709%	88%	80%	68%
Robot Components	-	65%	133%	37%	34%	34%
Other main business	-	-36%	31%	184%	10%	10%
Others	-	219%	354%	5%	5%	5%
Total Sales	-	147%	333%	60%	59%	52%
Sales mix	2023	2024	2025	2026E	2027E	2028E
Quadruped Robots	75%	59%	41%	33%	26%	20%
Humanoid Robots	2%	27%	51%	60%	68%	75%
Robot Components	17%	11%	6%	5%	4%	4%
Other main business	5%	1%	0%	1%	1%	0%
Others	1%	1%	1%	1%	1%	0%
Total Sales	100%	100%	100%	100%	100%	100%

Source: Unitree, BofA Global Research estimate

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In 2025, Unitree's selling & marketing/administrative/R&D expenses represented 8%/24%/9% of total revenue. Administrative expenses rose sharply to RMB358mn in 2Q25, mainly due to RMB349mn of share-based compensation recognized under the employee equity incentive plan. R&D expense increased 154% YoY and 15% QoQ to RMB63mn in 1Q26, then rose a further 151% YoY and 15% QoQ to RMB73mn. We expect the R&D expense ratio to exceed 11% in 2026-28, driven by accelerated investment in embodied AI models and motion-control algorithms – including higher spending on model training, data collection, and computing resources – as well as new robot product development. Looking ahead, we expect selling & marketing/administrative/R&D expenses to represent 6%/3%/12% of revenue by 2028 as HR sales ramp up.



Unitree's net profit rose 191% YoY to RMB278mn in 2025, driven by rapid shipment-led revenue growth and GPM expansion as the HR ramp improved the product mix. Excluding the RMB349mn impact of non-cash share-based compensation, adjusted net profit reached RMB591mn in 2025. In 1Q26, net profit declined 48% YoY amid higher R&D expense and slower revenue growth. In 2Q26, net profit increased 348% QoQ to RMB224mn (vs RMB128mn net loss in 2Q25 dragged by RMB349mn share-based payment expense), driven by 73% QoQ sales improvement in 2Q26. Looking ahead, we expect net profit to see an 84% CAGR over 2025-28, driven by a 57% revenue CAGR over the same period, led by HR shipment growth and operating leverage, partly offset by GPM normalization and higher R&D investment.

Exhibit 46: Unitree's P/L summary

We expect Unitree to deliver net profit CAGR of 84% over 2025-28

(RMB mn)	2023	2024	2025	2026E	2027E	2028E
Consolidated sales	159	393	1,699	2,713	4,312	6,552
Gross profit	71	225	1,027	1,494	2,274	3,275
Operating profit (clean EBIT)	-30	69	335	818	1,268	1,837
Impairment losses	-8	-10	-16	-16	-16	-16
Pre-tax profit	-21	101	368	808	1,378	1,955
Income tax	-10	5	90	78	165	235
Tax rate	47.9%	5.1%	24.5%	9.6%	12.0%	12.0%
Net income before minority	-11	95	278	730	1,212	1,720
Net income	-11	95	278	730	1,212	1,720
EBITDA	-23	78	358	992	1,647	2,379
Per share data (RMB)						
EPS	(0.03)	0.26	0.76	1.81	3.00	4.25
BVPS	0.82	3.52	6.44	22.87	25.87	30.13
Growth (%)						
Sales growth (%)		147%	333%	60%	59%	52%
Key ratio						
Gross margin	44.8%	57.2%	60.4%	55.1%	52.7%	50.0%
Operating margin (clean EBIT)	-18.9%	17.6%	19.7%	30.2%	29.4%	28.0%
Net margin	-7.0%	24.3%	16.4%	26.9%	28.1%	26.3%
EBITDA margin	-14.2%	19.8%	21.1%	36.6%	38.2%	36.3%
OPEX to sales	63.7%	39.6%	40.7%	24.9%	23.3%	21.9%

Source: Unitree, BofA Global Research estimate

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Exhibit 47: Unitree's quarterly P/L summary

We expect Unitree's revenue to grow 88%/55% YoY in 3Q/4Q26

(RMB mn)	1Q25	2Q25	3Q25	4Q25	2025	1Q26	2Q26	3Q26	4Q26	2026E
Revenue	251	525	392	532	1,699	423	729	738	822	2,713
Cost of goods sold	99	210	160	203	672	176	331	335	377	1,219
Gross profit	152	314	232	329	1,027	247	398	403	445	1,494
Promotion	23	24	29	65	141	113	51	51	53	269
G&A	6	358	17	18	400	19	23	30	33	105
R&D	25	29	36	55	145	63	73	74	82	292
Operating taxes and surcharges	1	2	1	3	7	2	3	3	3	11
Operating expenses	54	414	84	141	692	197	150	158	172	676
Operating income	98	-99	148	188	335	50	249	246	274	818
Pretax income	108	-93	156	197	368	49	252	240	267	808
Taxes or tax credit	13	34	19	25	90	-1	28	24	27	78
Profit before minority	96	-128	137	173	278	50	224	216	241	730
Minority interest	0	0	0	0	0	0	0	0	0	0
Net income	96	-128	137	173	278	50	224	216	241	730
Number of shares (mn)	404	404	404	404	404	404	404	404	404	404
Operating Ratios										
Gross margins	60.7%	59.9%	59.2%	61.8%	60.4%	58.4%	54.6%	54.6%	54.2%	55.1%
Promotion	9.0%	4.6%	7.4%	12.3%	8.3%	26.8%	7.0%	6.9%	6.5%	9.9%
G&A	2.3%	68.3%	4.5%	3.4%	23.5%	4.5%	3.2%	4.0%	4.0%	3.9%
R&D	9.9%	5.5%	9.3%	10.3%	8.5%	14.9%	10.0%	10.0%	10.0%	10.8%
Operating margin	39.2%	-18.9%	37.7%	35.3%	19.7%	11.9%	34.1%	33.3%	33.3%	30.2%
Pre-tax margin	43.1%	-17.8%	39.8%	37.1%	21.7%	11.6%	34.6%	32.5%	32.5%	29.8%
Tax rate	11.7%	-37.0%	11.9%	12.4%	24.5%	-2.1%	11.2%	10.0%	10.0%	9.6%
Net margin	38.1%	-24.3%	35.1%	32.5%	16.4%	11.8%	30.7%	29.2%	29.3%	26.9%
YoY %										
Net revenues	278.0%	667.8%	296.0%	234.2%	332.6%	68.5%	39.0%	88.4%	54.7%	59.7%
Gross profit	327.9%	741.7%	305.6%	247.2%	357.0%	62.2%	26.7%	74.0%	35.6%	45.5%
Operating income	9003.2%	-1884.3%	741.4%	319.0%	385.0%	-49.1%	N.M	66.3%	45.8%	144.5%
Net income	1554.6%	-1169.4%	424.6%	235.2%	191.4%	-47.7%	N.M	57.0%	39.2%	162.5%
QoQ %										
Net revenues	57.7%	109.1%	-25.3%	35.7%		-20.5%	72.5%	1.2%	11.4%	
Gross profit	61.0%	106.3%	-26.3%	41.7%		-24.8%	61.2%	1.3%	10.5%	
Operating income	119.8%	N.M	-248.9%	27.0%		-73.3%	396.3%	-1.2%	11.3%	
Net income	85.3%	N.M	-207.6%	25.9%		-71.1%	347.9%	-3.7%	11.6%	

Source: Unitree, BofA Global Research estimate

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Balance sheet and cash flow

Unitree's cash conversion cycle improved significantly from 283 days in 2023 to 56 days in 2025, primarily as inventory days fell from 328 to 138. The improvement was supported by faster inventory turnover amid rapid shipment growth in quadruped and humanoid robots, together with stronger supply-chain management. We expect the cash conversion cycle to remain stable at 54 days in 2026-28, supported by continued cash collection from growing HR sales, disciplined working-capital management, and efficient inventory control.

Exhibit 48: Unitree's cash conversion cycle

We expect Unitree's cash conversion cycle to maintain stable in 2025-28

	2023	2024	2025	2026F	2027F	2028F
AR days	24	14	9	9	9	9
Inventory days	328	239	138	135	135	135
AP days	69	116	91	90	90	90
Cash conversion cycle	283	138	56	54	54	54

Source: Unitree, BofA Global Research estimate

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Exhibit 49: Unitree's balance sheet

We expect Unitree's cash on hand to reach >RMB8bn at end-2026

(RMB mn)	2023	2024	2025	2026F	2027F	2028F
Total Assets	391	1,528	3,209	9,967	11,472	13,511
Current Assets	326	1,271	2,220	8,852	10,137	12,117
Cash & ST Investment	222	1,054	1,701	8,160	8,938	10,404
Inventories	79	141	368	534	974	1,450
Accounts Receivable	10	21	59	67	134	172
Others	15	55	91	91	91	91
Other Assets	65	257	989	1,115	1,336	1,394
LT Investments	0	0	0	0	0	0
Net fixed Assets	7	11	35	162	384	443
Others	58	246	954	953	952	951
Total Liabilities	92	247	604	715	1,008	1,326
Current Liabilities	85	238	556	667	960	1,278
Accounts Payable	17	90	245	356	649	967
ST Borrowings	16	5	16	16	16	16
Others	52	143	295	295	295	295
Long-term Liabilities	8	9	48	48	48	48
Long-term Debts	0	0	0	0	0	0
Others	8	9	48	48	48	48
Shareholders' Equity	299	1,281	2,605	9,252	10,464	12,185
Paid-in Capital	2	3	364	6,281	6,281	6,281
Capital Surplus	331	1,216	2,053	2,053	2,053	2,053
Retained Earnings	-34	61	166	897	2,109	3,829
Capital Adjustment	0	1	21	21	21	21
Minority interest	0	0	0	0	0	0

Source: Unitree, BofA Global Research estimate

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Unitree listed on the Shanghai Stock Exchange in August 2026, raising RMB6.1bn through its IPO by issuing 40 million shares at RMB150.8 per share. Net proceeds were RMB5.9bn after IPO fees, including sponsorship, underwriting, audit, and legal fees. Of the proceeds, RMB2.0bn is earmarked for embodied AI model R&D to advance foundation models, training infrastructure, and real-world datasets, thereby improving robots' task execution and operational intelligence. A further RMB1.1bn will be allocated to developing robotic hardware technologies, including high-power-density actuators, lightweight robot structures, and advanced motion-control systems. Furthermore, RMB1.7bn of the proceeds will be allocated to property, plant, and equipment (PP&E) investments, likely over the next 2-3 years. We estimate annual capex of RMB300-600mn. Accordingly, we forecast annual capex of RMB300-600mn over 2026-28, primarily for equipment procurement and site investment across the embodied AI model and hardware projects, R&D for new robot platforms, and construction of a robot manufacturing base.



Exhibit 50: Unitree's Capex plan

We expect Unitree's capex to be RMB300-600mn annually over 2026-28

(RMB mn)	2023	2024	2025	2026E	2027E	2028E
Capex	-5	-10	-50	-300	-600	-600
Operating cash flow	5	192	670	841	1,378	2,067

Source: Unitree, BofA Global Research estimate

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Exhibit 51: Unitree's cash flow statement

Unitree IPO in 2026 raised net proceeds of RMB5.9bn

(RMB mn)	2023	2024	2025	2026F	2027F	2028F
Operating Cash Flow	5	192	670	841	1,378	2,067
Net Profit	-11	95	278	730	1,212	1,720
Depreciation & Amortization	7	9	24	174	379	542
Change in Working Capital	13	76	-9	-63	-214	-196
Others	-4	12	377	0	0	0
Investment Cash Flow	78	-714	-472	-300	-600	-600
Net CAPEX	-5	-10	-50	-300	-600	-600
Change in LT Investment	0	0	0	0	0	0
Change in Other Assets	83	-704	-422	0	0	0
Adj. Free Cash Flow*	83	-521	198	541	778	1,467
Financing Cash Flow	3	860	675	5,917	0	0
Change in Share Capital	0	885	695	5,917	0	0
Net Change in Debt	0	-11	11	0	0	0
Change in Other LT Liab.	3	-14	-31	0	0	0
Other cash item	3	4	-23	0	0	0
Net Cash Flow	88	343	850	6,459	778	1,467

Source: Unitree, BofA Global Research estimate (*Note: *Adjusted FCF includes the impact of changes in short-term investments, which we consider a use of cash when assessing underlying free cash flow generation)

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Company background

Founded in August 2016, Unitree Robotics (Unitree) is a global leader in high-performance, general-purpose robotics, specializing in humanoid and quadruped robots, core robotic components, and embodied AI. Leveraging full-stack, in-house R&D capabilities across robotics algorithms, intelligent systems, and key components – including motors, reducers, dexterous robotic hands, LiDAR, and sensors – the company has enhanced robot integration, whole-body mobility, and dexterity. Its products are widely used by research institutions, enterprises, and developers worldwide in applications spanning scientific research, industrial inspection, public services, and consumer-oriented scenarios.

Following the IPO, founder Mr. Wang Xingxing directly holds 21.44% of the Company's shares and c.30% combining direct and indirect ownership. Through the Company's weighted voting rights structure, Mr. Wang Xingxing controls c.65% of the total voting power, thereby maintaining effective control of the Company post-listing.

Exhibit 52: Unitree’s history

Unitree was established in 2016 and listed on the Shanghai Stock Exchange in 2026

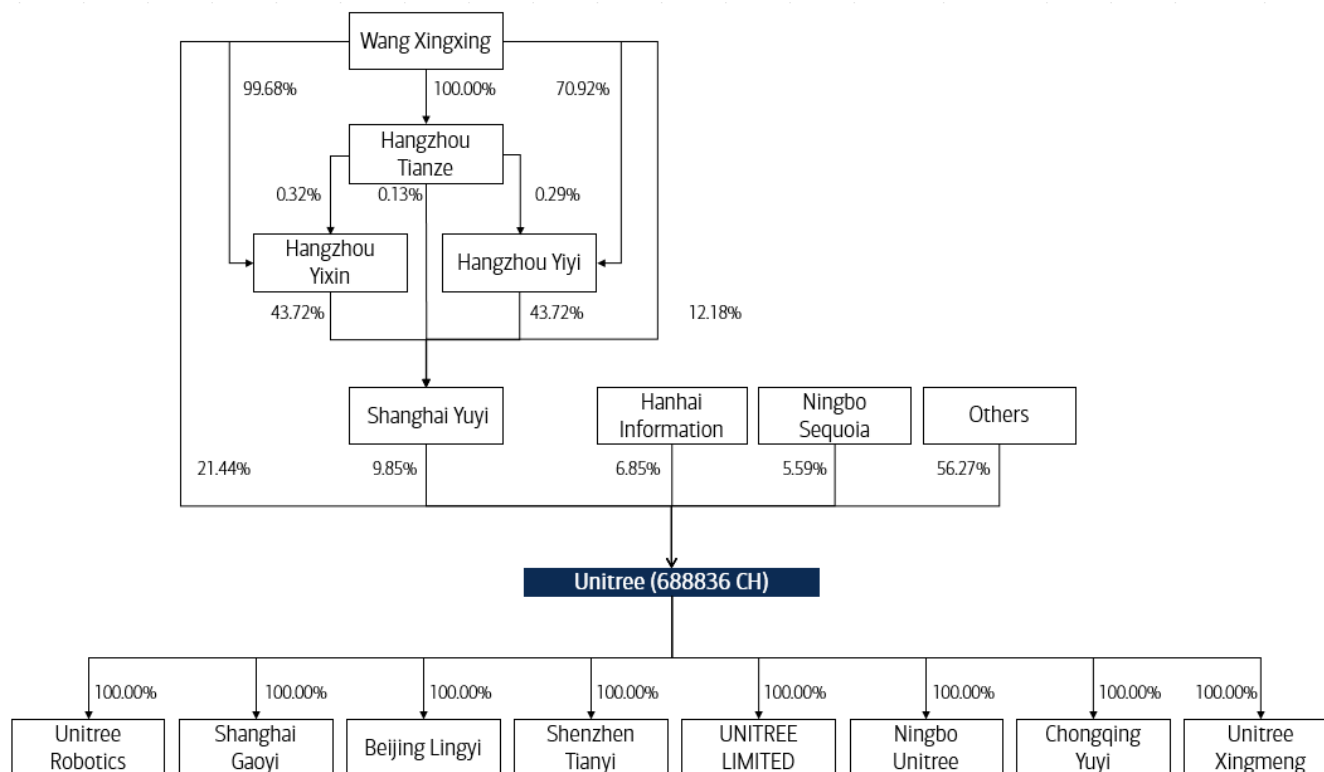
Year	Key milestone
2016	Unitree Robotics was founded in Hangzhou by Wang Xingxing, initially focusing on the development and n of high-performance quadruped robots.
2017	Launched Laikago, its first publicly available quadruped robot, marking the beginning of the company’s commercial product portfolio.
2019	Introduced AlienGo, expanding its quadruped portfolio toward higher-payload and more demanding applications.
2020	Launched A1, a lightweight and agile quadruped platform targeting research and developer applications.
2021	Launched Go1, broadening its reach into the mass-market consumer and research segments; Unitree robots also appeared on the CCTV Spring Festival Gala.
2022	Expanded into industrial quadrupeds with B1 and showcased 109 quadruped robots at the opening ceremony of the Beijing Winter Olympics.
2023	Launched Go2 and industrial-grade B2, and entered the humanoid robot market with H1, extending its portfolio from quadruped robots to general-purpose humanoid platforms.
2024	Launched G1, a medium-sized humanoid robot, accelerating the accessibility and n of humanoid robots.
2025	Achieved rapid shipment growth across humanoid and quadruped robots, while continuing to expand its embodied AI capabilities.
2025	H1 robots appeared on the CCTV Spring Festival Gala.
2025	Launched its second full-sized general-purpose humanoid robot, H2.
2026	Listed on the Shanghai Stock Exchange’s STAR Market.

Source: Company prospectus



Exhibit 53: Unitree's shareholding structure

Unitree founder Mr. Wang Xingxing directly holds a 21.44% stake, with around 30% of the shareholding combining direct and indirect ownerships



Source: Company prospectus

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Exhibit 54: Unitree's management team

Mr. Wang Xingxing is the Chairman, General Manager, and CTO of Unitree

Name	Age	Position	Background
Wang Xingxing	36	Chairman, General Manager & CTO	Mr. Wang has served as chairman, general manager and CTO of the company since 2016. He has over 15 years of robotics R&D experience. During his master's studies, he pioneered the development of XDog, the world's first quadruped robot powered by low-cost outer-rotor brushless motors, and was named to TIME's "The 100 Most Influential People in AI 2025". He holds a Master's degree from Shanghai University.
Yang Zhiyu	35	Director, Head of mechanical structure	Mr. Yang joined the company in 2016 and currently serves as director and head of mechanical structure. He has nearly 10 years of experience in robotic mechanical system development and has led the mechanical design, core component development, and overall system design of multiple generations of Unitree's flagship robotic products. He holds a Master's degree from Zhejiang University.
Zhang Yangguang	33	Director, Head of algorithms and software	Mr. Zhang joined the company in 2018 and currently serves as director and head of algorithms and software. He has nearly 10 years of experience in motion control algorithms for legged and humanoid robots. He has led the development and iteration of motion control technologies across its core product portfolio. Prior to joining the company, he worked as an algorithm engineer at Surgnova Healthcare Technologies (Beijing) Co., Ltd. from April to October 2017. Before that, he worked at Qinu (Beijing) Technology Co., Ltd. from 2015 to 2017. He holds a bachelor's degree from Nankai University.
Wang Feng	44	CFO	Mr. Wang joined the company in 2021 and currently serves as CFO. Prior to joining the company, he served as CFO at Zhejiang Shuanglin Environment Co., Ltd. from May 2015 to March 2021 and at Hangzhou Feizhu Technology Co., Ltd. from April 2014 to May 2015. He previously worked as finance manager at Ningbo Xintai Machinery Co., Ltd. from June 2009 to April 2014 and as assistant director at Chun'an Zhenye Tax Agent Firm Co., Ltd. from July 2006 to May 2009.
Fu Fenghua	38	Board Secretary	Mr. Fu joined the company in 2025 and currently serves as board secretary. Prior to joining the company, he worked at Zhejiang Hangke Technology Co., Ltd. from June 2018 to May 2025, where he held various positions including deputy financial director, financial director and board secretary, and deputy general manager. He previously worked at Pan-China Certified Public Accountants LLP from 2010 to 2018.

Source: Company prospectus

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Risks

Slower-than-expected humanoid robot adoption

We expect Unitree to ship 17.0k/31.7k/52.7k HRs and 32.8k/44.0k/51.7k quadruped robots in 2026/27/28, driven by expanding applications and improving affordability that support broader adoption. However, shipment and revenue growth could be negatively affected if HR development is slower than expected because of: (1) slower progress in embodied AI and foundation models; (2) weaker-than-expected advances in motion control and dexterous manipulation; (3) slower reductions in production and component costs; or (4) delayed adoption of humanoid robot applications.

Intensifying competition in humanoid and quadruped robots

Unitree has established a leading position in the humanoid and quadruped robot markets through its technological expertise and early-mover advantage. However, competition continues to intensify as more robotics start-ups, technology companies, and automotive OEMs enter the industry. If Unitree cannot maintain its market leadership in the coming years, its shipment growth and GPM could be negatively affected.

Shortages of critical components and materials

HR development relies on several critical components, including: (1) high-performance chips, which are dominated by NVIDIA; (2) planetary roller screws used in HR linear actuators; (3) sensors used in electronic skin and actuators; and (4) NdFeB magnets used in high-performance motors. If demand on HR outpaces the supply growth of key components, a shortage of these critical components and materials could negatively affect shipments of Unitree's humanoid robots.

Availability of funding for HR development

The HR industry remains highly capital- and technology-intensive, requiring substantial and sustained investment in hardware platforms, embodied AI models, algorithm development, and computing infrastructure. Although Unitree has demonstrated strong growth momentum, its future expansion will depend on adequate funding for R&D and commercialization. A shortage of capital could negatively affect Unitree's operations and further R&D.

FCC restrictions could limit future access to the U.S. market

Unitree's existing humanoid and quadruped robot models, including the G1, H2, R1, Go2, B2, and A2, have received Federal Communications Commission (FCC) certification and remain eligible for sale in the U.S. market. However, the FCC's latest restrictions on foreign-produced advanced robotic devices may make it more difficult for future models to obtain the authorization required to enter the U.S. market. Further regulatory tightening could also affect previously certified products. If Unitree cannot maintain access to the US market, its overseas expansion, shipment growth, and profitability could be negatively affected.



Exhibit 55: Abbreviations

Abbreviations used in the report

Abbreviation	Full name	Abbreviation	Full name
3D	Three-dimensional	LLM	Large language model
AI	Artificial intelligence	OTA	Over-the-air
ASP	Average selling price	R&D	Research and development
BOM	Bill of materials	SoC	System on chip
CCID	China Center for Information Industry Development	STAR	Science, Technology and Innovation Registration-based Board
DoF	Degree of freedom	UIO	Units in ownership
DRL	Deep reinforcement learning	UMI	Universal manipulation interface
HR	Humanoid robot	VLA	Vision-language-action
IMU	Inertial measurement units	VTLA	Vision-tactile-language-action
IPO	Initial public offering	WAM	World action models
LiDAR	Light detection and ranging	WMA	World-model-action

Source: BofA Global Research

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Price objective basis & risk

Unitree / Yushu Technology (XYAJF)

Our PO of RMB350 is derived from the DCF valuation methodology, reflecting Unitree's direct exposure to humanoid robot, n, superior sales/earnings growth profile, leadership in China's HR market, and execution track record. We assume 12.0% WACC and 8% FCF perpetual growth.

Upside risks: faster-than-expected humanoid robot adoption, stronger demand from enterprise customers, accelerated n of embodied AI capabilities, faster product iteration, and greater-than-expected cost reductions supporting margin upside.

Downside risks: slower-than-expected development of humanoid robot, intensifying competition, lack of critical components and materials, availability of financial resources.

Analyst Certification

I, Ming Hsun Lee, CFA, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

APR - Industrials Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Aboitiz Equity Ventures, Inc.	ABOIF	AEV PM	Julian Roxas
	AirTAC	ATCXF	1590 TT	Ming Hsun Lee, CFA
	Anhui Yingliu	XANHf	603308 CH	Fiona Liang
	Ayala Corporation	AYYLF	AC PM	Russ Toribio
	Centre Testing International	XCTEF	300012 CH	Ming Hsun Lee, CFA
	Daifuku	DAIUF	6383 JP	Kenjin Hotta
	Daikin Industries	DKILF	6367 JP	Kenjin Hotta
	Daikin Industries	DKILY	DKILY US	Kenjin Hotta
	Dtech Technology	XDTXF	301377 CH	Summer Wang, CFA
	Dtech Technology	XNISF	1377 HK	Summer Wang, CFA
	Envicool	XPTIF	002837 CH	Yibing Xia
	GeekPlus	BGTcf	2590 HK	Summer Wang, CFA
	Goldwind	XJNGF	2208 HK	Gary Tsang
	GTCP	GTHOF	GTCP PM	Russ Toribio
	Guangdong Songfa	XDZGF	603268 CH	Amy Han
	Guizhou Aviation	XNSZF	688239 CH	Fiona Liang
	Hainan Jinpan	XKZNF	688676 CH	Matty Zhao
	Hangcha Group	XXCLF	603298 CH	Summer Wang, CFA
	Hangzhou Great Star	XNZLF	002444 CH	Fiona Liang
	Hankuk Carbon	HKUCF	017960 KS	KJ Hwang
	Han's Laser	XHQTF	002008 CH	Summer Wang, CFA
	Hanwha Aerospace	SSASF	012450 KS	KJ Hwang
	HD Hyundai Electric	XPELF	267260 KS	Sun Jung Lee
	HD Hyundai Heavy Industries	XHZXF	329180 KS	KJ Hwang
	Himile	XMSVF	002595 CH	Fiona Liang
	Hiwin Technologies	HITeF	2049 TT	Ming Hsun Lee, CFA
	Huaming Equipment	XKTIF	002270 CH	Matty Zhao
	IHI	IHICF	7013 JP	Kenjin Hotta
	Inovance Technology	XMYRF	300124 CH	Ming Hsun Lee, CFA
	Japan Steel Works	JPSWF	5631 JP	Yuri Nishizaki
	Jiangsu Hengli Hydraulic Co., Ltd	XHDCF	601100 CH	Ming Hsun Lee, CFA
	Kawasaki Heavy Industries	KWHIF	7012 JP	Kenjin Hotta
	Keyence	KYCCF	6861 JP	Kenjin Hotta
	Korea Aerospace	XAQHF	047810 KS	KJ Hwang
	LIG Defense & Aerospace	XLQUF	079550 KS	KJ Hwang
	LS Electric	LGXSF	010120 KS	Sun Jung Lee
	Makita	MKEWF	6586 JP	Kenjin Hotta



APR - Industrials Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	Mingyang Smart Energy	XLARF	601615 CH	Charlotte Xia
	Mitsubishi Heavy Industries	MHVF	7011 JP	Kenjin Hotta
	Mitsui E&S	MIESF	7003 JP	Yuri Nishizaki
	MNC Solution	XZXR	484870 KS	Dongho Shin
	Nabtesco	NCTKF	6268 JP	Kenjin Hotta
	Namura Shipbuilding	NMRSF	7014 JP	Yuri Nishizaki
	NARI Technology	XLODF	600406 CH	Matty Zhao
	OMRON Corporation	OMRNF	6645 JP	Kenjin Hotta
	OMRON Corporation	OMRNY	OMRNY US	Kenjin Hotta
	RoboTechnik	XYICF	300757 CH	Matty Zhao
	Samsung Heavy Industries	SMSHF	010140 KS	KJ Hwang
	Sany Heavy Industry Co., Ltd	SHVF	600031 CH	Ming Hsun Lee, CFA
	Sany Heavy Industry Co., Ltd	XHEVF	6031 HK	Ming Hsun Lee, CFA
	Sany International	SNYYF	631 HK	Summer Wang, CFA
	SATS Limited	SPASF	SATS SP	Nathan Gee, CFA
	Shenling	XSRLF	301018 CH	Yibing Xia
	Shenzhen Megmeet	XXMLF	002851 CH	Fiona Liang
	Sieyuan Electric	XUJF	002028 CH	Matty Zhao
	Sinexcel	XSBEF	300693 CH	Summer Wang, CFA
	Sinotruk Hong Kong Ltd	SHKLF	3808 HK	Fiona Liang
	SM Investments Corporation	SVTMF	SM PM	Russ Toribio
	SMC	SMECF	6273 JP	Kenjin Hotta
	SMC	SMCAF	SMCAF US	Kenjin Hotta
	ST Engineering	SGGKF	STE SP	Jessie Lo
	Suzhou Maxwell Technologies	XSMTF	300751 CH	Gary Tsang
	TGOOD	XQGDF	300001 CH	Ming Hsun Lee, CFA
	THK	THKLF	6481 JP	Kenjin Hotta
	UBTECH	UBTRF	9880 HK	Ming Hsun Lee, CFA
	Weichai Power Co. Ltd	WEICF	2338 HK	Fiona Liang
	Weichai Power Co. Ltd	XWPEF	000338 CH	Fiona Liang
	XCMG Construction Machinery Co Ltd	XVPZF	000425 CH	Ming Hsun Lee, CFA
	XJ Electric	XJECF	000400 CH	Matty Zhao
	Yangzijiang Shipbuilding	YSHLF	YZJSGD SP	Amy Han
	Yaskawa Electric	YASKF	6506 JP	Kenjin Hotta
	Yuchai International	CYD	CYD US	Fiona Liang
	Zhejiang Supcon Technology	XDXFF	688777 CH	Ming Hsun Lee, CFA
	Zoomlion Heavy Industry	XLRYF	000157 CH	Ming Hsun Lee, CFA
	Zoomlion Heavy Industry	ZLIOF	1157 HK	Ming Hsun Lee, CFA

NEUTRAL

	Anhui Heli	AHUHF	600761 CH	Summer Wang, CFA
	D&L Industries, Inc.	DALQF	DNL PM	Philip Albert Felix
	Dongguan Yiheda Automation	XDGF	301029 CH	Ming Hsun Lee, CFA
	Fanuc	FANUF	6954 JP	Kenjin Hotta
	Fanuc	FANUY	FANUY US	Kenjin Hotta
	Kubota Corp	KUBTF	6326 JP	Kenjin Hotta
	Kubota Corp	KUBTY	KUBTY US	Kenjin Hotta
	Neway Valve	XNNVF	603699 CH	Summer Wang, CFA
	Zhuzhou CRRC Times Electric	ZHUZF	3898 HK	Ming Hsun Lee, CFA
	Zhuzhou CRRC Times Electric	XWCTF	688187 CH	Ming Hsun Lee, CFA

UNDERPERFORM

	Alliance Global Group, Inc.	ALGGF	AGI PM	Russ Toribio
	CKD	CKDOF	6407 JP	Kenjin Hotta
	ESTUN Automation	XNJNF	002747 CH	Ming Hsun Lee, CFA
	ESTUN Automation	XZMTF	2715 HK	Ming Hsun Lee, CFA
	Goldwind	XJGWF	002202 CH	Gary Tsang
	Hitachi CM	HTCMF	6305 JP	Kenjin Hotta
	Hongfa Technology	XQXAF	600885 CH	Fiona Liang
	Komatsu	KMTUF	6301 JP	Kenjin Hotta
	Komatsu	KMTUY	KMTUY US	Kenjin Hotta
	L.K. Technology	XBQXF	558 HK	Joey Yang, CFA
	Leader Harmonious Drive Systems	XLHNF	688017 CH	Ming Hsun Lee, CFA
	LT Group, Inc.	LTGIF	LTG PM	Philip Albert Felix
	OPT Machine Vision	XOTVF	688686 CH	Summer Wang, CFA
	SCG Packaging PCL	XAFLF	SCGP TB	Komsun Suksumrun



APR - Industrials Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	Shanghai Bochu Electronic	XWUTF	688188 CH	Fiona Liang
	Shenzhen SC New Energy	XENYF	300724 CH	Ming Hsun Lee, CFA
	SK Bend	SUKWF	014620 KS	KJ Hwang
	Unitree / Yushu Technology	XYAJF	688836 CH	Ming Hsun Lee, CFA

iQmethodSM Measures Definitions**Business Performance**

Return On Capital Employed

Return On Equity

Operating Margin

Earnings Growth

Free Cash Flow

Numerator

NOPAT = (EBIT + Interest Income) × (1 – Tax Rate) + Goodwill Amortization

Net Income

Operating Profit

Expected 5 Year CAGR From Latest Actual

Cash Flow From Operations – Total Capex

Denominator

Total Assets – Current Liabilities + ST Debt + Accumulated Goodwill

Amortization

Shareholders' Equity

Sales

N/A

N/A

Quality of Earnings

Cash Realization Ratio

Asset Replacement Ratio

Tax Rate

Net Debt-To-Equity Ratio

Interest Cover

Numerator

Cash Flow From Operations

Capex

Tax Charge

Net Debt = Total Debt – Cash & Equivalents

EBIT

Denominator

Net Income

Depreciation

Pre-Tax Income

Total Equity

Interest Expense

Valuation Toolkit

Price / Earnings Ratio

Price / Book Value

Dividend Yield

Free Cash Flow Yield

Enterprise Value / Sales

Numerator

Current Share Price

Current Share Price

Annualised Declared Cash Dividend

Cash Flow From Operations – Total Capex

EV = Current Share Price × Current Shares + Minority Equity + Net Debt +

Other LT Liabilities

Enterprise Value

Denominator

Diluted Earnings Per Share (Basis As Specified)

Shareholders' Equity / Current Basic Shares

Current Share Price

Market Cap = Current Share Price × Current Basic Shares

Sales

Basic EBIT + Depreciation + Amortization

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Equity Investment Rating Distribution: Electrical Equipment Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	18	66.67%	Buy	10	55.56%
Hold	3	11.11%	Hold	2	66.67%
Sell	6	22.22%	Sell	1	16.67%

Equity Investment Rating Distribution: Engineering Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	5	83.33%	Buy	1	20.00%
Hold	1	16.67%	Hold	1	100.00%
Sell	0	0.00%	Sell	0	0.00%

Equity Investment Rating Distribution: Industrials/Multi-Industry Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	68	62.39%	Buy	36	52.94%
Hold	15	13.76%	Hold	7	46.67%
Sell	26	23.85%	Sell	12	46.15%

Equity Investment Rating Distribution: Machinery/Diversified Manufacturing Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	39	59.09%	Buy	18	46.15%
Hold	12	18.18%	Hold	9	75.00%
Sell	15	22.73%	Sell	9	60.00%

Equity Investment Rating Distribution: Global Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	1987	56.11%	Buy	1190	59.89%
Hold	797	22.51%	Hold	496	62.23%
Sell	757	21.38%	Sell	391	51.65%

^{R1} Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster ^{R2}
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Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

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